

CLAT Express - September 2026: Study Notes

Chapters, topics and sub-topics with key facts, memory aids and mind-map outlines.

International Relations and News

Houthi Rebels Capture Strategic Port City Mokha in Yemen

Capture of Mokha and Red Sea stakes

Iran-backed Houthi forces seized the Red Sea port city of Mokha in Taiz Governorate, along with Perim and Zuqar islands. Mokha lies about 80 km from the Bab el-Mandeb Strait, which links the Red Sea with the Gulf of Aden, so Houthi control of the area threatens shipping routes between Asia and Europe. Government forces withdrew south towards Dhubab, while Houthi forces advanced towards the Hanish Islands.

- Houthis, an Iran-backed group, seized Mokha port city in Taiz Governorate.
- Houthis also seized Perim and Zuqar Islands in Yemen.
- Mokha is located about 80 km from the Bab el-Mandeb Strait.
- Bab el-Mandeb Strait links the Red Sea with the Gulf of Aden.
- After Mokha fell, government forces moved south towards Dhubab.
- Houthi forces advanced towards the Hanish Islands in the Red Sea.
- Saudi Arabia shut its East-West Oil Pipeline after drone strikes from Iraq.
- Earlier Houthi attacks on Red Sea shipping forced some vessels onto longer routes around Africa.
- Renewed fighting caused civilian displacement and disrupted local supply lines.

Mind-map outline

Mokha capture

- Territory taken
 - Mokha port city
 - Perim and Zuqar Islands
- Strategic position
 - 80 km from Bab el-Mandeb
 - Red Sea to Gulf of Aden link
- Shipping impact
 - Ships forced onto routes around Africa
 - Asia-Europe trade threatened

Houthi movement: identity and rise

The Houthis, officially known as Ansar Allah, are mainly Zaidi Shia Muslims and members of the Houthi tribe, and they oppose Israel and the US. Their founder, Hussein al-Houthi, was killed by Yemeni forces in Saada in 2004, which sparked the Houthi insurgency and the Saada Wars. The Houthis later seized Saada and captured Sanaa in 2014, forcing President Hadi to flee abroad in 2015.

- The Houthi movement is officially known as Ansar Allah (Supporters of God).
- Houthis are primarily composed of Zaidi Shia Muslims and members of the Houthi tribe.
- Founder: Zaidi religious leader Hussein al-Houthi; current leader: Abdul-Malik al-Houthi.
- Houthis began as an opposition movement to President Ali Abdullah Saleh in the 2000s.
- Founder was killed by Yemeni forces in Saada in 2004, sparking the Houthi insurgency.
- Saada Wars ran from 2004 to 2010.
- The 2011 Yemen Uprising forced Saleh to transfer power to his deputy, Abdrabbuh Mansour Hadi.
- Houthis seized Saada and captured Sanaa in 2014, forcing Hadi to flee abroad in 2015.
- Houthis claimed responsibility for a missile attack on Riyadh's King Khalid International Airport in 2017.
- Houthis ideology: opposition to Israel and the US; part of the Iranian-led Axis of Resistance.

Mind-map outline

Houthi movement

- Identity
 - Ansar Allah (Supporters of God)

- Zaidi Shia Muslims
- Members of the Houthi tribe
- Leaders
 - Hussein al-Houthi: founder
 - Abdul-Malik al-Houthi: current leader
- Key events
 - 2004: founder killed, Saada Wars
 - 2011: Saleh hands power to Hadi
 - 2014: Sanaa captured
 - 2015: Hadi flees abroad
- Stances
 - Opposes Israel and the US
 - Part of Axis of Resistance

Civil war and the 2018 peace deals

After the Houthi takeover of Sanaa, Saudi Arabia intervened militarily from 2015 to restore the internationally recognised government. The Stockholm Agreement, reached in Sweden on 13 December 2018, has three main components and was endorsed by the UN Security Council under resolution 2451. The Hodeidah ceasefire, effective from midnight on 17 December 2018, required the Houthis to withdraw from the ports and Hodeidah City under UN-chaired supervision.

- The Saudi-led intervention began in 2015 to reinstate the internationally recognised government.
- The Stockholm Agreement was agreed in Sweden on 13 December 2018 at UN-mediated talks.
- The Stockholm Agreement has three main components: Hudaydah Agreement, Prisoner Exchange Agreement and Taiz Agreement.
- The UN Security Council endorsed the Stockholm Agreement under resolution 2451 (2018).
- The Hodeidah ceasefire took effect from midnight on 17 December 2018.
- The ceasefire covered Hodeidah City and the ports of Hodeidah, Ras Isa and Saleef.
- Under the agreement, the Houthis withdraw from the ports and from Hodeidah City.
- A UN-chaired committee including both sides oversees the withdrawal of forces.
- Yemeni ports fall under local forces, who send port revenues to the country's central bank.

Mnemonic - Stockholm Agreement Components: Hot Pizza Tastes

H = Hudaydah Agreement; P = Prisoner Exchange Agreement; T = Taiz Agreement

Mnemonic - Hodeidah Ceasefire Ports: Hot Rice Served

H = Hodeidah port; R = Ras Isa port; S = Saleef port

Mind-map outline

Civil war and peace deals

- Saudi-led intervention
 - Began in 2015
 - Aim: restore recognised government
- Stockholm Agreement
 - Agreed 13 Dec 2018 in Sweden
 - Endorsed under resolution 2451
 - Hudaydah Agreement
 - Prisoner Exchange Agreement
 - Taiz Agreement
- Hodeidah ceasefire
 - Effective from midnight 17 Dec 2018
 - Houthis withdraw from ports and city
 - UN-chaired withdrawal committee
 - Port revenues to central bank

North-South divide and southern separatism

Yemen existed as two separate states until 1990, when the North and South merged under Ali Abdullah Saleh. Power stayed heavily centralised after unification, which fuelled southern resentment and led to an attempted secession and brief civil conflict in 1994. The Southern Movement (Al-Hirak) formed in 2007, and the UAE-backed Southern Transitional Council now seeks autonomy or independence for the south.

- Until 1990, Yemen existed as two countries: the Yemen Arab Republic (North) and the People's Democratic Republic of Yemen (South).
- North Yemen was supported by the US and Saudi Arabia; South Yemen was backed by the Soviet Union.
- In 1990 the two states merged to form the Republic of Yemen.
- Ali Abdullah Saleh had governed North Yemen since 1978 and became President of the unified state.
- Centralised power after unification created resentment in the south.
- By 1994 southern leaders attempted secession, leading to a brief civil conflict.
- The Southern Movement (Al-Hirak) formed in 2007 and demands greater autonomy or independence.
- The Southern Transitional Council is backed by the UAE and has expanded influence in Hadramout and Mahra.
- A pro-government military bloc opposes the STC's separatist agenda and receives backing from Saudi Arabia.

Mind-map outline

Yemen North-South divide

- Before 1990
 - Two separate states
 - North backed by US and Saudi Arabia
 - South backed by Soviet Union
- Unification in 1990
 - Saleh becomes President
 - Centralised power fuels resentment
- Southern separatism
 - 1994 secession attempt
 - 2007 Al-Hirak formed
 - STC backed by UAE
- Current rivals
 - Pro-government bloc opposes STC
 - Hadramout Tribal Alliance ties
 - Saudi backing for pro-government bloc

Mukalla strike and UAE troop withdrawal

Saudi Arabia carried out airstrikes on Mukalla Port in southern Yemen after a consignment from Fujairah in the UAE arrived there, which Riyadh alleged contained arms for separatist groups. The UAE rejected the allegation, saying the vehicles were meant exclusively for its own forces. The UAE then pulled its troops out, citing security risks, a counter-terrorism-only mandate and the view that Yemen's future should be decided by Yemenis.

- Saudi Arabia carried out airstrikes on Mukalla Port in southern Yemen.
- The strike followed the arrival of a consignment from Fujairah (UAE).
- Riyadh alleged the consignment contained arms meant for southern separatist groups.
- The UAE said the shipment consisted of vehicles meant exclusively for UAE forces.
- The UAE termed the claims of an arms transfer to separatists unfounded.
- Security concern: escalating airstrikes raised risks to UAE personnel in the region.
- Limited mandate: the UAE's military presence in Yemen is confined to counter-terrorism operations.
- Political messaging: Yemen's future political structure should be determined by Yemenis themselves.

Mind-map outline

Mukalla strike and UAE withdrawal

- Trigger
 - Saudi airstrikes on Mukalla Port
 - Consignment from Fujairah alleged arms

- UAE response
 - Vehicles meant for UAE forces only
 - Arms claims termed unfounded
- Withdrawal reasons
 - Risks to UAE personnel
 - Counter-terrorism mandate only
 - Yemenis to decide future

Indus Waters Treaty and India's rejection of the Hague ruling

Treaty basics and river allocation

The Indus Waters Treaty, 1960, is a bilateral water-sharing agreement between India and Pakistan that was facilitated by the World Bank. It allocates the Eastern rivers to India and the Western rivers primarily to Pakistan. India keeps limited, conditional rights on the Western rivers, including run-of-river power generation.

- Signed on 19 September 1960, after eight years of negotiations
- Bilateral water-sharing agreement between India and Pakistan
- Facilitated by the World Bank
- Eastern rivers: Ravi, Beas and Sutlej, with exclusive and unrestricted use for India
- Western rivers: Indus, Jhelum and Chenab, allocated primarily to Pakistan
- India's rights on Western rivers are limited and subject to treaty conditions
- Permitted uses on Western rivers include domestic use, limited agricultural use and run-of-river hydroelectric generation
- India can build run-of-river hydro projects with minimum storage for conservation and flood management

Mnemonic - Western rivers to Pakistan: I Jump Cheerfully

I = Indus is a Western river allocated primarily to Pakistan; J = Jhelum is a Western river allocated primarily to Pakistan; C = Chenab is a Western river allocated primarily to Pakistan

Mind-map outline

Treaty basics and allocation

- Signed 19 Sept 1960 after eight years
- Facilitated by the World Bank
- Western rivers to Pakistan
 - Indus, Jhelum, Chenab
- Eastern rivers to India
 - Ravi, Beas, Sutlej
- India's limited Western rights
 - Run-of-river power
 - Domestic and non-consumptive use

Three-tier dispute-resolution mechanism

The treaty provides a three-tiered route for disputes, starting with the Permanent Indus Commission for bilateral discussion. Technical differences that the Commission cannot resolve may go to a Neutral Expert, while formal legal disputes may be referred to a Court of Arbitration. The Court of Arbitration is constituted under Annexure G of the treaty.

- The treaty provides a three-tiered dispute mechanism
- Permanent Indus Commission is the first institutional mechanism and provides a forum for bilateral discussion and information exchange
- The Permanent Indus Commission meets at least once a year
- Neutral Expert handles technical differences the Commission cannot resolve
- The Neutral Expert is appointed through a mechanism involving the World Bank
- The Neutral Expert route primarily addresses technical questions
- Court of Arbitration handles formal legal disputes referred under the treaty
- The Court of Arbitration is constituted under Annexure G and consists of 5 members

Mnemonic - Dispute ladder order: Come Eat Cake

Come = Permanent Indus Commission is the first tier for bilateral discussion; Eat = Neutral Expert takes

technical differences the Commission cannot resolve; Cake = Court of Arbitration takes formal legal disputes

Mind-map outline

Three-tier dispute mechanism

- Permanent Indus Commission
 - Bilateral discussion and information exchange
 - Meets at least once a year
- Neutral Expert
 - Technical differences
 - Appointed via World Bank mechanism
- Court of Arbitration
 - Formal legal disputes
 - Constituted under Annexure G
 - 5 members

Court of Arbitration's findings

The Court of Arbitration held that the treaty continues to apply and that India's decision to place it in abeyance was not permissible. It said the treaty does not deal with terrorism or use of force, so cross-border terrorism was not accepted as a material breach. The tribunal also imposed interim limits on construction at the Ratle project.

- The treaty continues in force until terminated by a duly ratified treaty (Article XII(4))
- Article III(1) obliges India to let flow the waters of the Western rivers to Pakistan
- Placing the treaty in abeyance was held not permissible under the treaty or international law
- The Vienna Convention on the Law of Treaties 1969 was applied in the ruling
- The treaty does not provide India a unilateral mechanism to suspend obligations
- The treaty contains no provisions on terrorism or use of force
- Interim limits bar concreting of specified parts of the Ratle dam wall and power intake beyond prescribed levels
- The Ratle restrictions last until 90 days after the Neutral Expert's final determination

Mind-map outline

Court of Arbitration holdings

- Treaty continues in force
 - Article XII(4)
 - Article III(1): let flow to Pakistan
- Abeyance not permissible
 - Vienna Convention 1969
- Terrorism not a treaty subject
 - No provisions on terrorism or force
- Ratle interim limits
 - Until 90 days after Neutral Expert decision

India's objection to the tribunal

India rejects the ruling, saying the tribunal was constituted by the World Bank contrary to the treaty and has no jurisdiction over its sovereign decisions. India argues the treaty sets a graded, sequential route that ends with a court only after the Neutral Expert step. Pakistan and India initiated parallel proceedings in 2016, and the World Bank paused and later resumed the two processes.

- India has rejected the ruling and has neither recognised the tribunal's legitimacy nor participated in its proceedings
- India's Ministry of External Affairs says the Court of Arbitration was constituted by the World Bank contrary to the treaty
- India's abeyance decision after the April 2025 Pahalgam terror attack remains valid according to New Delhi
- India says the tribunal has no jurisdiction over its sovereign decisions or infrastructure projects on the western rivers
- India's position: question, then Permanent Indus Commission, then Neutral Expert, then legal dispute or Court of Arbitration

- In 2016, Pakistan initiated proceedings seeking a Court of Arbitration
- In 2016, India requested appointment of a Neutral Expert to examine technical aspects
- In 2016, the World Bank paused both the arbitration and Neutral Expert processes
- In 2022, the World Bank allowed both processes to move forward
- India took part in Neutral Expert proceedings but not in Court of Arbitration proceedings

Mind-map outline

India's objection to the tribunal

- Tribunal improperly constituted
 - World Bank constituted it, per MEA
- Graded, sequential route
 - Commission, then Neutral Expert, then court
- Parallel proceedings
 - 2016: Pakistan sought Court of Arbitration
 - 2016: World Bank paused both processes
 - 2022: both processes resumed

Indus river system and water-sharing principles

The Indus is a trans-boundary river and the longest river in Pakistan, and its main tributaries have distinct sources in Tibet, Kashmir and Himachal Pradesh. The booklet lists three international instruments on shared water: the Helsinki Rules of 1966, the Helsinki Convention of 1992 and the UN Watercourses Convention of 1997. India, China and Pakistan have not signed the UN Watercourses Convention.

- The Indus is a trans-boundary river and the longest river in Pakistan
- The Indus originates at the Bokhar Chu Glacier near Mt Kailash, Mansarovar, Tibet
- The Jhelum rises at the Verinag Spring in the Kashmir Valley and joins the Chenab at Trimmu
- The Chenab originates near Bara Lacha Pass in Lahul-Spiti (HP) and joins the Satluj at Panchnad
- The Ravi rises in the Kullu Hills near Rohtang Pass and joins the Chenab a little above Rangpur
- The Beas originates near Rohtang Pass and joins the Satluj at Harike
- The Satluj rises from the Manasarovar-Rakas Lakes in Tibet and joins the Indus a little above Mithankot
- Helsinki Rules 1966 were adopted by the International Law Association to regulate water use in an international drainage basin
- Helsinki Convention 1992 provides a framework for preventing and controlling water pollution across national borders
- UN Watercourses Convention 1997 codified law on non-navigational uses of international watercourses; India, China and Pakistan have not signed it

Mnemonic - International water instruments in order: Use Pollution Codes

Use = Helsinki Rules 1966 regulate use of water in an international drainage basin; Pollution = Helsinki Convention 1992 covers preventing and controlling water pollution across borders; Codes = UN Watercourses Convention 1997 codified law on non-navigational uses

Mind-map outline

Indus river system and principles

- Source points
 - Indus: Bokhar Chu Glacier, Tibet
 - Jhelum: Verinag Spring, Kashmir
 - Chenab: Bara Lacha Pass area
 - Ravi and Beas: Rohtang Pass area
 - Satluj: Manasarovar-Rakas Lakes
- International principles
 - Helsinki Rules 1966: water use
 - Helsinki Convention 1992: pollution
 - UN Watercourses Convention 1997

Cardiff Pact and UK Federal Challenge

Devolution and the asymmetric UK framework

Devolution transfers powers from the central government to subnational institutions, while the centre keeps ultimate constitutional authority. In the UK, Scotland, Wales and Northern Ireland have devolved institutions, but the extent and nature of their powers differ, so the arrangement is asymmetric rather than uniform. The UK Treasury uses the Barnett Formula to set changes in block grant funding for these administrations.

- Devolution means transfer of powers from central government to subnational institutions
- Ultimate constitutional authority is retained at the centre
- Devolved institutions in the UK: Scotland, Wales and Northern Ireland
- Extent and nature of devolved powers differ among the three
- UK devolution is asymmetric rather than uniform
- Scotland Act 1998 established the Scottish Parliament and devolved substantial domestic powers
- Government of Wales Act 1998 established the Welsh Legislature, now known as the Senedd
- Barnett Formula is used by the UK Treasury to set block grant changes
- UK Internal Market Act 2020 regulates the domestic UK market after Brexit
- Devolved governments criticised the 2020 Act as reducing their regulatory autonomy

Mind-map outline

Devolution and asymmetry

- Transfer of powers from centre
 - Centre retains ultimate authority
- Asymmetric, not uniform
 - Scotland Act 1998: Scottish Parliament
 - Government of Wales Act 1998: Senedd
 - Barnett Formula: block grant changes
- UK Internal Market Act 2020
 - Regulates domestic market after Brexit
 - Criticised as cutting devolved autonomy

Historical formation of the Union

The UK's constituent nations joined the union at different points. Wales was incorporated into the English legal and political system through the Act of Union of 1536, while later Acts of Union brought Scotland (1707) and Ireland (1801) into the state. After most of Ireland became independent in 1922, six counties remained within the UK as Northern Ireland.

- Wales was incorporated into the English legal and political system; Act of Union in 1536
- Acts of Union, 1707 united the English and Scottish Parliaments into the Kingdom of Great Britain
- Act of Union, 1801 brought Ireland into the United Kingdom
- Anglo-Irish War (1919-21) preceded Irish independence
- In 1922 most of Ireland became independent
- Six counties formed Northern Ireland within the UK
- The state is now the United Kingdom of Great Britain and Northern Ireland

Mind-map outline

Formation of the Union

- Wales: Act of Union 1536
- 1707: English and Scottish Parliaments united
- 1801: Ireland brought into UK
- Irish independence
 - Anglo-Irish War 1919-21
 - 1922: most of Ireland independent
 - Six counties remained in UK

Good Friday Agreement and Northern Ireland

The Good Friday (Belfast) Agreement, signed on 10 April 1998, ended the conflict known as The Troubles in Northern Ireland. It set up power-sharing government, allowed reunification with Ireland subject to a referendum, and let citizens hold Irish or British nationality or both. After the 2022 deadlock over government formation, a new deal was published as the command paper Safeguarding the Union.

- Signed on 10 April 1998 at Belfast; ended a 30-year-old civil war known as The Troubles
- Power-sharing government formed on the wishes of Republicans and Unionists, who share governance equally
- People can seek reunification with Ireland at any time, subject to a referendum
- Citizens can hold Irish or British nationality or both
- Agreement abolished border checks and encouraged freedom of movement across UK and Ireland
- A border poll on Irish reunification can be called by the UK Secretary of State for Northern Ireland when conditions show majority support
- After Brexit, Northern Ireland shared a land border with the Republic of Ireland, an EU member
- Border checks between Great Britain and Northern Ireland angered Unionists
- The DUP leader refused to allow government formation after the May 2022 Northern Ireland polls
- A new deal was published as a command paper called Safeguarding the Union

Mnemonic - Good Friday Agreement's three aspects: SRN: Share, Reunite, Nationality

S = Government formed on both sides' wishes, sharing governance equally; R = Reunification with Ireland possible any time, subject to a referendum; N = Citizens can hold Irish or British nationality or both

Mind-map outline

Good Friday Agreement 1998

- Signed 10 April 1998 at Belfast
- Three main aspects
 - Power-sharing government
 - Reunification referendum any time
 - Irish or British nationality or both
- Border and aftermath
 - Border poll: Secretary of State can call
 - 2022 government formation refused
 - Safeguarding the Union command paper

Cardiff Pact and the UK federal challenge

The Cardiff Pact is a political Memorandum of Understanding signed in Cardiff by leaders from Scotland, Wales and Northern Ireland, and it is not a legally binding treaty. It asks Westminster to prepare, plan and facilitate constitutional change and affirms each nation's right to hold independence referendums. The UK government rejects new referendums, relying on a 2022 Supreme Court ruling that the Scottish Parliament cannot legislate for one without Westminster's consent.

- Cardiff Pact is an inter-party accord signed in Cardiff (Wales)
- Signatories include John Swinney (SNP), Rhun ap Iorwerth (Plaid Cymru) and Michelle O'Neill (Sinn Féin Vice President)
- Mary Lou McDonald (Sinn Féin President) also attended
- The pact is a political Memorandum of Understanding, not a legally binding treaty
- Core objectives: prepare, plan and facilitate constitutional change; affirm right to hold independence referendums; coordinate policies on economy, energy and European relations
- Scotland and Wales would seek EU membership as new states
- Northern Ireland would rejoin the EU through unification with the Republic of Ireland
- All three devolved administrations are now led by pro-independence parties
- UK government rejects new independence referendums without appropriate constitutional and legal authority
- 2022 UK Supreme Court ruling (Reference by the Lord Advocate): Scottish Parliament cannot unilaterally legislate for an independence referendum without Westminster's consent

Mind-map outline

Cardiff Pact

- Signed in Cardiff, Wales
- Political MoU, not a treaty
- Objectives
 - Prepare, plan and facilitate change
 - Affirm independence referendum right
 - Coordinate economy, energy, Europe
- UK government: no new referendums yet

Lindsey O. Graham Sanctioning Russia and Iran Act 2026

Passage and timeline

The United States enacted the Lindsey O. Graham Sanctioning Russia and Iran Act of 2026 after House passage and Senate approval. The House voted 262 to 159 on 16 September, and the President signed the Act into law on 18 September 2026. The measure matters for India because India remains one of the major buyers of Russian crude.

- House of Representatives passed the legislation 262-159 on 16 Sep
- The legislation followed Senate approval
- Presidential signing into law on 18 Sep 2026
- Act aims primarily at increasing economic pressure on Russia over the Ukraine War
- Act also extends and strengthens certain sanctions against Iran
- India is significant because it remains one of the major buyers of Russian crude

Mind-map outline

Passage of the Act

- House vote 262-159 on 16 Sep
- Senate approval preceded signing
- Signed into law 18 Sep 2026

Key provisions

The Act authorises tariffs of up to 100% on the five largest purchasers of Russian oil and natural gas, a substantial reduction from the 500% original proposal. It also authorises tariffs of up to 500% on Russian-origin goods, subject to specified exceptions. A 180-day review cycle for relevant purchasers and defined conditions for ending the sanctions framework complete its core design.

- Tariffs of up to 100% on the 5 largest purchasers of Russian oil and natural gas
- 100% ceiling is a substantial reduction from the original 500% proposal
- Tariffs also target the Top 5 facilitators of Russian oil sanctions evasion
- USTR reassesses relevant purchasers under a 180-day review cycle
- Tariffs of up to 500% on Russian-origin goods, subject to specified exceptions
- Iran Sanctions Act of 1996 is extended through 2031
- Gas exception: importing less than 15% of natural gas from Russia and taking significant steps to reduce such imports
- Framework may be affected if Russia signs an accepted peace agreement, ceases military hostilities and ends activities undermining the Ukrainian Government

Mnemonic - Conditions for ending measures: PCE: Peace, Cease, End

P = Russia signs a peace agreement accepted by the internationally recognised Government of Ukraine;
C = Russia ceases military hostilities; E = Russia ends activities aimed at undermining the Ukrainian Government

Mind-map outline

Key provisions

- Tariffs
 - Up to 100% on five largest buyers
 - Up to 500% on Russian-origin goods
- Review
 - USTR reassesses purchasers
 - 180-day cycle
- Exceptions

- Gas imports below 15% from Russia
- Ending conditions
 - Accepted peace agreement
 - Ceasing military hostilities
 - Ending undermining of Ukrainian government

Legal permanence and US trade tools

The earlier 25% punitive tariff on India was based on an Executive Order, and in February 2026 the U.S. Supreme Court struck down the broader tariff regime that included it. The new measure has been passed by Congress, signed and converted into federal law, which gives it greater legal permanence. A presidential waiver under the Act requires formal justification to Congress.

- Earlier 25% punitive tariff on India was based on an Executive Order
- In February 2026 the U.S. Supreme Court struck down the broader tariff regime
- New measure was passed by Congress, signed and converted into federal law
- Federal law gives the new measure greater legal permanence than an executive tariff
- Presidential waiver requires formal justification to Congress
- Section 301 of the Trade Act, 1974 lets the USTR investigate foreign trade practices
- Section 232 of the Trade Expansion Act, 1962 restricts imports seen as a threat to national security
- United States imposed tariffs of 50% on steel, aluminium, copper and certain related products
- India faces a 10% tariff linked to a U.S. investigation into goods produced using forced labour

Mnemonic - US trade statutes: Probe in '74, Block in '62

Probe '74 = Section 301 of the Trade Act, 1974 lets the USTR investigate foreign trade practices; Block '62 = Section 232 of the Trade Expansion Act, 1962 restricts imports seen as a threat to national security

Mind-map outline

Legal permanence and US tools

- Earlier 25% tariff on India
 - Based on an Executive Order
 - Regime struck down, Feb 2026
- New Act
 - Passed by Congress, signed into law
 - Waiver needs formal justification to Congress
- Other US tools
 - Section 301, Trade Act 1974
 - Section 232, Trade Expansion Act 1962
 - 50% on steel, aluminium, copper

Global and India implications

The measure illustrates growing use of secondary sanctions and tariff instruments to influence the behaviour of third countries. It could change global oil trade patterns as major buyers diversify suppliers and reconfigure energy contracts. For India, Russia is the top oil supplier, so potential tariffs complicate bilateral trade while India balances strategic autonomy with its energy interests.

- Measure illustrates growing use of secondary sanctions and tariff instruments
- Major buyers may diversify suppliers and reconfigure energy contracts
- Buyers may increase purchases before potential tariff measures
- Buyers may develop alternative payment and logistics arrangements
- Russia is India's top oil supplier
- Potential tariffs complicate India-U.S. bilateral trade and economic ties
- Higher import bills drive up transport, logistics, manufacturing and consumer prices
- Discounted crude improves economics, though replacing large volumes is logistically challenging
- Balancing relationships with major powers while protecting national energy interests

Mnemonic - Buyer responses: DRIP

D = Diversify suppliers; R = Reconfigure energy contracts; I = Increase purchases before potential tariff measures; P = Develop alternative payment and logistics arrangements

Mind-map outline

Global and India implications

- Global oil trade
 - Diversify suppliers
 - Reconfigure energy contracts
 - Alternative payment and logistics
- India
 - Russia is top oil supplier
 - Friction in India-US trade
 - Balancing strategic autonomy
- Spillovers
 - Higher import bills, rising consumer prices
 - Discounted crude improves economics

India Rejects Pakistan-China Boundary Joint Commission

Pakistan-China Boundary Joint Commission

Pakistan and China have set up a Boundary Joint Commission that traces back to their 2013 boundary management agreement and was operationalised in September 2026. India rejects the body because it treats the territory involved as its own and does not recognise the 1963 agreement under which Shaksgam Valley passed to China. The Commission sits alongside China-Pakistan connectivity links such as the Khunjerab crossing and CPEC, which India opposes.

- Pakistan-China Boundary Joint Commission was operationalised in Sep 2026; inaugural meeting held in Islamabad.
- The Commission originates from the 2013 Pakistan-China Agreement on Boundary Management System.
- Article 45 of the 2013 agreement provides for a Joint Commission to supervise implementation of border management arrangements.
- Commission is intended to facilitate coordination on border management, joint surveys, trade flows and people-to-people connectivity.
- 1963 China-Pakistan Boundary Agreement was signed on 2 March 1963.
- Under the 1963 agreement, Pakistan transferred approx 5,180 sq km of Shaksgam Valley (Trans-Karakoram Tract) to China.
- India has never recognised the 1963 Sino-Pakistan Boundary Agreement and considers it illegal and invalid.
- India argues Pakistan has no locus standi over territory India regards as under Pakistan's illegal occupation.
- Khunjerab Crossing/Corridor is the principal overland connection between China and Pakistan.
- CPEC, a major component of China's Belt and Road Initiative, connects Kashgar in Xinjiang with Gwadar Port via Pakistan; India opposes it because parts pass through Pakistan-occupied Kashmir.

Mind-map outline

Pakistan-China Boundary Joint Commission

- Origin
 - 2013 Agreement on Boundary Management System
 - Article 45 provides for the Commission
- Purpose
 - Border management coordination
 - Joint surveys and trade flows
 - People-to-people connectivity
- India's position
 - No legal basis in territory concerned
 - Pakistan lacks locus standi
- Shaksgam Valley
 - 1963 agreement signed 2 March 1963
 - About 5,180 sq km passed to China

Sino-Indian Border: States, UTs and Sectors

India shares a 3,488 km boundary with China along the Himalayan range, and this border runs through one Union Territory and four states. Ladakh has the longest border segment with China among all Indian states and UTs, while Arunachal Pradesh has the second longest. For management purposes the Line of Actual Control is divided into Eastern, Middle and Western sectors, each grouping specific states and UTs.

- India shares a 3,488 km border with China along the Himalayan range.
- The border runs through 5 entities: 1 UT (Ladakh) and 4 states (Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh).
- Ladakh shares the longest border segment with China among all Indian states and UTs.
- Arunachal Pradesh has the 2nd longest border segment with China.
- Himachal Pradesh borders China and forms part of the Middle Sector.
- Uttarakhand borders China and Nepal, among other states.
- Sikkim borders China, Nepal and Bhutan.
- LAC sectors: Eastern (Arunachal Pradesh and Sikkim), Middle (Uttarakhand and Himachal Pradesh), Western (Ladakh).
- Panchsheel Agreement was signed by Nehru and Zhou Enlai, with both sides pledging to respect each other's sovereignty and territorial integrity.

Mind-map outline

Sino-Indian Border

- 3,488 km along Himalayas
 - 1 UT and 4 states
- Longest and second-longest segments
 - Ladakh: longest
 - Arunachal Pradesh: second longest
- LAC sectors
 - Eastern: Arunachal Pradesh, Sikkim
 - Middle: Uttarakhand, Himachal Pradesh
 - Western: Ladakh

Disputed Lines and Flashpoints

The India-China boundary is disputed in two main areas: Aksai Chin in the west and Arunachal Pradesh in the east. Each side relies on different historical lines, and China rejects the McMahon Line that India recognises. The disputes have produced repeated clashes, from the 1962 war to the 2017 Doklam standoff and the 2020 Galwan Valley clash.

- India considers Aksai Chin part of Ladakh and relies on the Johnson Line; China administers it as part of Xinjiang Autonomous Region and relies on the MacDonald Line.
- India recognises the McMahon Line, demarcated at the 1914 Simla Convention, as the legal boundary; China rejects it and claims Arunachal Pradesh as South Tibet.
- China contends that Tibet was not a sovereign state and could not conclude treaties, which is its reason for rejecting the Shimla agreement.
- The term Line of Actual Control was introduced by Chinese Premier Zhou Enlai in 1959.
- The LAC came into practical existence after the 1962 Sino-Indian War.
- PLA invaded India on 20 Oct 1962; China declared a unilateral ceasefire on 19 Nov 1962 but retained Aksai Chin.
- 1967 Nathu La conflict and 1975 Tulung La ambush, in which four Indian soldiers were killed.
- 2017 Doklam plateau standoff lasted 73 days after India sent troops to stop China building a road.
- 15 Jun 2020 Galwan Valley clash: at least 20 Indian soldiers killed, the most serious clash in 50 years.

Mnemonic - Flashpoints in order: Naughty Tigers Dance Gracefully

N = Nathu La conflict, 1967; T = Tulung La ambush, 1975; D = Doklam plateau standoff, 2017; G = Galwan Valley clash, 2020

Mind-map outline

Disputed Lines and Flashpoints

- Western: Aksai Chin
- India: Johnson Line, part of Ladakh

- China: administers via Xinjiang
- Eastern: Arunachal Pradesh
 - India: McMahon Line, 1914
 - China: South Tibet claim
- Flashpoints
 - 1962 war: ceasefire 19 Nov 1962
 - 1967 Nathu La, 1975 Tulung La
 - 2017 Doklam: 73 days
 - 2020 Galwan Valley clash

Border Management Mechanisms and Agreements

India and China manage the LAC through a layered set of mechanisms and agreements rather than a final settlement of the boundary. The 2003 Special Representatives process is the highest political-level channel for a settlement, while the 2012 Working Mechanism coordinates responses to LAC incidents. Agreements from 1993 to 2013 added rules on peace, military activity near the LAC and face-to-face encounters between troops.

- Special Representatives on Boundary Question (2003) is the highest political-level mechanism for a final political settlement of the 3,488 km dispute.
- Working Mechanism for Consultation and Coordination (2012) is a diplomatic and military channel for LAC incidents and has no mandate to settle the boundary.
- Corps Commander-level meetings pursue tactical disengagement, de-escalation and buffer-zone arrangements during major standoffs.
- Border Personnel Meeting Points are designated locations where local military commanders address ground-level friction.
- 1993 Agreement for Maintenance of Peace and Tranquillity along the LAC recognised the LAC; it was signed during PM Narasimha Rao's visit to China.
- 1996 Confidence-Building Measures agreement restricted military activities and use of weapons near the LAC; signed during Jiang Zemin's visit to India.
- 2005 CBM Protocol set procedures for face-to-face encounters between troops.
- 2013 Border Defence Cooperation Agreement strengthened communication and coordination between border forces.

Mnemonic - Border management bodies: Sip Warm Chai, Biscuits

S = Special Representatives (2003): highest political-level mechanism; W = Working Mechanism (2012): diplomatic and military channel; C = Corps Commander-level meetings: tactical disengagement; B = Border Personnel Meeting Points: ground-level friction

Mind-map outline

Managing the LAC

- Mechanisms
 - Special Representatives (2003)
 - Working Mechanism (2012)
 - Corps Commander-level meetings
 - Border Personnel Meeting Points
- Agreements
 - 1993 peace and tranquillity
 - 1996 CBM military agreement
 - 2005 CBM Protocol
 - 2013 Border Defence Cooperation

Map Projections: From Mercator to Equal Earth

Why flat maps must distort

The Earth is an oblate spheroid, so its curved surface cannot be shown on a flat map without distortion. A projection converts the 3D surface into a 2D representation, and no single flat map keeps area, shape, distance and direction exact at once. Conformal, equal-area and equidistant projections each preserve one property, while compromise projections balance distortion across all four.

- Earth is an oblate spheroid rather than a perfect sphere.
- The oblate shape is due to Earth's rotation and uneven distribution of mass.
- A projection converts Earth's 3D curved surface into a 2D representation.
- Some degree of distortion is inevitably introduced by any projection.
- The four properties are area, shape, distance and direction.
- Conformal projections preserve angles and local shapes (e.g. Mercator).
- Equal-area projections preserve relative areas (e.g. Equal Earth).
- Equidistant projections preserve distance.
- Compromise projections balance distortions but preserve none of the properties exactly.
- Navigation and direction use conformal projections.

Mind-map outline

Flat map distortion

- Earth shape: oblate spheroid
 - Caused by rotation
 - Caused by uneven mass
- Projection: 3D to 2D
 - Distortion is unavoidable
- Properties traded off
 - Area
 - Shape
 - Distance
 - Direction

Mercator projection and its distortion

The Mercator projection was introduced in 1569 by Gerardus Mercator and was primarily designed to assist maritime navigation. It is a cylindrical and conformal projection, so it keeps local shapes and shows rhumb lines as straight lines. The cost is that landmasses far from the Equator look much larger than they are, and Africa appears similar in size to Greenland.

- Introduced in 1569 by Gerardus Mercator, a Flemish geographer and cartographer.
- Type: cylindrical and conformal projection.
- Primarily designed to assist maritime navigation.
- Earth's surface is conceptually projected onto a cylinder around the globe, then unfolded flat.
- Stretching of parallels is matched exactly with stretching of meridians, which preserves shapes.
- Landmasses far from the Equator appear much larger than they are.
- Africa appears similar in size to Greenland, though Greenland is in reality 14 times smaller.
- Rhumb lines appear as straight lines, which makes the projection useful for navigation.
- Does not preserve relative area.

Mnemonic - Mercator's features: Mercator Makes Routes Straight

Mercator = Introduced in 1569 by Gerardus Mercator; Makes = Designed primarily to assist maritime navigation; Routes = Rhumb lines are useful for navigation; Straight = Rhumb lines appear as straight lines on the Mercator map

Mind-map outline

Mercator projection (1569)

- Cylindrical, conformal
- Designed for navigation
 - Rhumb lines appear straight
- Distortion near the poles
 - High-latitude landmasses enlarged
 - Africa looks like Greenland in size

Equal Earth and the Correct the Map resolution

Equal Earth is an equal-area, pseudo-cylindrical world map projection created in 2018 to show landmass sizes accurately without severe shape distortion. The UN General Assembly adopted a 'Correct the Map' resolution, led by Togo, that promotes equal-area projections to address Mercator distortions. The resolution is not binding, so whether governments and classrooms adopt the new map remains to be seen.

- Equal Earth is an equal-area, pseudo-cylindrical world map projection, created in 2018.
- Designed to display the entire globe while preserving relative landmass sizes accurately.
- Avoids the severe, aesthetically jarring shape distortions common in earlier equal-area maps such as Gall-Peters.
- The UN General Assembly adopted the 'Correct the Map' resolution promoting equal-area map projections.
- Togo led the resolution through Foreign Minister Robert Dussey, under the #CorrectTheMap campaign.
- The resolution was supported by 164 countries, including India.
- The United States voted against it and six countries abstained.
- The African Union endorsed it in August 2025 and the Caribbean Community backed it too.
- Correct the Map is run by two advocacy groups, Africa No Filter and Speak Up Africa.
- India backed the resolution while drawing a firm line on depiction of Jammu and Kashmir and Ladakh; UN resolutions are not binding.

Mind-map outline

Equal Earth and Correct the Map

- Equal Earth (2018)
 - Equal-area, pseudo-cylindrical
 - Accurate relative landmass sizes
- UN resolution
 - Led by Togo
 - 164 countries supported
 - United States voted against
 - Not binding

Other projections and mapping in India

Gall-Peters, first presented in 1855 and popularised in 1973, preserves area but stretches or squashes shapes, while Robinson (1963) and Winkel Tripel (1921) are compromise projections. Winkel Tripel replaced Robinson in National Geographic Society maps in 1998. In India, the Survey of India, established in 1767, verifies international borders on published maps, and the Criminal Law Amendment Act (1961) penalises maps that misrepresent India's borders.

- The Gall-Peters projection was first presented by Scottish clergyman James Gall in 1855.
- Arno Peters popularised the Gall-Peters projection in 1973.
- Gall-Peters preserves area accurately but makes landmasses look stretched or squashed.
- The Robinson projection (1963) was devised by Arthur H. Robinson to balance size and shape.
- Robinson is a compromise projection, with meridians curving gently towards the Poles.
- The Winkel Tripel projection (1921) was created by German cartographer Oswald Winkel.
- Winkel Tripel aims to reduce distortion across area, distance and direction.
- In 1998, Winkel Tripel replaced Robinson in National Geographic Society maps, followed by schools, textbooks and the World Bank.
- The Survey of India, established in 1767, verifies international borders on published maps.
- Criminal Law Amendment Act (1961), Section 2, penalises publication and distribution of maps that question India's territorial integrity or misrepresent its borders.

Mnemonic - Chronology of three projections: Good Wine Rocks

G = Gall-Peters, first presented in 1855; W = Winkel Tripel, created in 1921; R = Robinson, devised in 1963

Mind-map outline

Other projections and India's maps

- Gall-Peters (1855)

- Presented by James Gall
- Popularised by Arno Peters, 1973
- Robinson (1963)
 - Balances size and shape
 - Meridians curve towards Poles
- Winkel Tripel (1921)
 - Replaced Robinson in National Geographic maps, 1998
- India's mapping rules
 - Survey of India, established 1767
 - Criminal Law Amendment Act 1961, Section 2

National News and Issues

Removal of Chief Election Commissioner

Constitutional Status and Powers of the ECI

The Election Commission of India is a permanent, independent constitutional body under Articles 324 to 329 (Part XV), established in 1950. It administers elections to Parliament, State Legislatures, the President and the Vice President, and has been a three-member body since 1993. Its powers are administrative, advisory and quasi-judicial, and the Supreme Court in *Mohinder Singh Gill* (1977) treated the case as a cornerstone for its plenary powers.

- Permanent, all-India, quasi-judicial and independent constitutional body
- Constitutional basis: Articles 324-329 (Part XV)
- Ensures free and fair elections for Parliament, State Legislatures, President and Vice President
- Three members since 1993: one CEC and two ECs
- Administrative power: defining constituencies, preparing voter lists, scheduling elections
- Advisory power: advises the President and Governors, guides during emergencies
- Quasi-judicial power: resolves disputes over party recognition, allocates election symbols
- *Mohinder Singh Gill v. CEC* (1977): cornerstone case on ECI's plenary powers under Article 324
- Where laws are silent (Articles 327 and 328), ECI can act independently to safeguard free and fair elections
- ECI can order a re-poll if the purity of the electoral process is compromised; judicial review usually comes after the election

Mind-map outline

ECI: constitutional status

- Basis
 - Articles 324-329 (Part XV)
 - Established 1950
- Composition
 - Three members since 1993
 - One CEC and two ECs
- Powers
 - Administrative: constituencies, voter lists
 - Advisory: President and Governors
 - Quasi-judicial: party recognition, symbols
- Judicial principles
 - *Mohinder Singh Gill* (1977)
 - Re-poll if purity compromised
 - Review usually after elections

Appointment, Tenure and Removal of CEC and ECs

The CEC and Other Election Commissioners Act, 2023 replaced the 1991 Act and provides for appointment by the President on the recommendation of a Selection Committee, following the Supreme Court's directive in *Anoop Baranwal vs Union of India* (2023). The CEC and ECs enjoy Supreme Court Judge-level salary and benefits. The CEC serves for six years or until 65 years of age, whichever is earlier, and is removed through the same process as a Supreme Court Judge.

- *Anoop Baranwal vs Union of India* (2023): five-judge bench directed appointment on the advice of a Selection Committee
- Earlier, the President made appointments on the advice of the Prime Minister
- The 2023 Act replaced the Election Commission (Conditions of Service of Election Commissioners and Transaction of Business) Act, 1991 on conditions of service and transaction of business
- Selection Committee: Prime Minister, a Union Cabinet Minister, and Leader of Opposition in Lok Sabha
- Search Committee, headed by the Minister of Law and Justice, proposes a panel of five names
- Selection Committee can consider names beyond the shortlist and can function despite vacancies
- Tenure: six years or 65 years of age, whichever is earlier; not eligible for re-appointment
- An EC appointed as CEC: combined term as EC and CEC must not exceed six years
- CEC and ECs enjoy Supreme Court Judge-level salary and benefits
- CEC is removed by the same process as a Supreme Court Judge

Mnemonic - Selection Committee members: PCL picks the Commissioners

P = Prime Minister; C = Union Cabinet Minister; L = Leader of Opposition (or largest opposition party) in Lok Sabha

Mind-map outline

CEC and ECs: appointment and removal

- Appointment
 - *Anoop Baranwal* (2023) directive
 - Selection Committee under 2023 Act
 - Search Committee proposes five names
- Tenure
 - Six years or age 65
 - Not eligible for re-appointment
 - EC-to-CEC combined cap: six years
- Status and removal
 - Supreme Court Judge-level salary
 - CEC removed like an SC Judge

Electoral Rolls, SIR and the Right to Vote

An electoral roll is the official list of persons eligible to vote in a jurisdiction, and being listed on it is a legal prerequisite for casting a vote. Rolls can be revised intensively, summarily or in special cases. The Special Intensive Revision (SIR) is a focused, time-bound house-to-house verification by Booth Level Officers, and courts have held that deletion from rolls cannot rest on mere suspicion.

- Electoral roll: official list of individuals eligible to vote in a given election within a jurisdiction
- Being on the electoral roll is a legal prerequisite for casting a vote in India
- Publishing and updating rolls: ECI, state's Chief Electoral Officer, and state election commission
- Article 325: no individual excluded from electoral rolls on grounds of religion, race, caste or sex
- Intensive revision: complete fresh preparation of rolls through house-to-house enumeration
- Summary revision: routine annual updating of existing rolls
- SIR: house-to-house verification by BLOs to add missing voters, remove ineligible ones and correct entries
- *Lal Babu Hussein vs Electoral Registration Officer* (1995): guidelines allowing deletion on mere suspicion struck down
- *Kuldip Nayar vs Union of India* (2006): right to vote is statutory, not fundamental
- Bihar SIR: those not in the 2003 rolls had to submit any of 11 listed documents to prove birth date or place

Mnemonic - Types of electoral roll revision: Fresh, Annual, Exceptional

Fresh = Intensive revision: fresh preparation of rolls; Annual = Summary revision: routine annual updating; Exceptional = Special revision: undertaken in exceptional cases

Mind-map outline

Electoral rolls, SIR, right to vote

- Electoral roll
 - Official list of eligible voters
 - Prerequisite for casting a vote
- Revision types
 - Intensive: fresh house-to-house roll
 - Summary: routine annual update
 - Special: exceptional cases
- SIR
 - Time-bound BLO verification
 - Add, remove and correct entries
 - Excludes non-citizens
- Right to vote
 - Article 326: adult suffrage
 - Statutory right (Kuldip Nayar 2006)

EVM and VVPAT

The Electronic Voting Machine (EVM) records votes in place of ballot papers and boxes, and was first used in Paravur, Kerala, in 1982. The VVPAT prints a slip that lets voters verify their vote; the slip stays visible for seven seconds before dropping into a sealed box. In 2019 the Supreme Court ruled that VVPAT slips of five EVMs in every constituency should be counted instead of just one.

- EVM: electronic device used to record votes in place of ballot papers and boxes
- First used in Paravur Assembly Constituency, Kerala, in 1982
- Developed by Bharat Electronics Ltd (Bangalore) and Electronic Corporation of India Ltd (Hyderabad)
- Two parts: Control Unit (with Polling Officer) and Balloting Unit (in voting booth)
- VVPAT slip shows the serial number, candidate name and symbol
- Slip is visible through a transparent window for 7 seconds, then is cut and falls into a sealed drop box
- VVPAT machines can be accessed by Polling Officers only
- VVPAT first introduced in the Noksen Assembly bye-election, Nagaland, in 2013
- VVPAT used in all constituencies in the 2019 Lok Sabha elections
- 2019 Supreme Court ruling: VVPAT slips of 5 EVMs per constituency to be counted instead of 1

Mnemonic - VVPAT slip: Seven seconds to see, then drop

Seven = Slip stays visible through the window for 7 seconds; See = Slip shows serial number, candidate name and symbol; Drop = Slip is cut and falls into a sealed drop box

Mind-map outline

EVM and VVPAT

- EVM
 - Replaced ballot papers and boxes
 - First used: Paravur, Kerala, 1982
 - Control Unit and Balloting Unit
- VVPAT
 - Slip visible for 7 seconds
 - Slip drops into sealed box
 - First used: Noksen, Nagaland, 2013
- Counting
 - 2019 SC: 5 EVMs per constituency
 - VVPAT in all constituencies (2019 LS)

Polity & Governance

SEMICON India 2026

Why Semiconductors Matter

Semiconductors are foundational to modern technology and are used in automobiles, healthcare, defence, aerospace, telecommunications and consumer electronics. Chips with billions of transistors power high-performance computing and artificial intelligence, within a global electronics industry valued at approximately USD 4.3 trillion. COVID-19 exposed the risks of concentrated global supply chains, which has pushed countries to build domestic capabilities.

- Semiconductors are used in automobiles, healthcare, defence, aerospace, telecommunications and consumer electronics.
- Availability of semiconductor components directly influences the resilience of digital and industrial supply chains.
- Modern chips can contain billions of transistors within extremely small areas.
- Leading-edge processors based on 3-nanometre technology can integrate tens of billions of transistors.
- Such capabilities support high-performance computing, artificial intelligence and advanced digital applications.
- The global electronics industry is valued at approximately USD 4.3 trillion.
- COVID-19 exposed vulnerabilities arising from concentration in global semiconductor supply chains.
- Countries have sought to develop domestic semiconductor capabilities and diversify sources.
- Materials with conductivity between conductors and insulators, such as silicon and germanium, are semiconductors.

Mind-map outline

Why Semiconductors Matter

- Used in
 - Automobiles, healthcare, defence
 - Aerospace, telecom, consumer electronics
- Technology
 - Billions of transistors per chip
 - 3-nm processors: tens of billions
- Economy
 - Electronics industry: about USD 4.3 trillion
- Supply chains
 - COVID-19 exposed concentration risk
 - Countries seek domestic capability

SEMICON India Conference and Programme 1.0

SEMICON India is an annual flagship conference and exhibition that showcases India's semiconductor capabilities, and the 2026 edition is the fifth. PM Modi inaugurated it at Yashobhoomi, New Delhi, under the theme 'Silicon to Systems: Building the Ecosystem'. The Semicon India Programme, launched in 2021, aims to establish a complete semiconductor ecosystem in India, with focus on chip design, fabrication, advanced packaging, talent development, and research and innovation.

- SEMICON India is an annual flagship semiconductor conference and exhibition.
- The first edition of SEMICON India was held in Bengaluru in 2022.
- The 2026 edition of SEMICON India is the fifth edition.
- PM Modi inaugurated SEMICON India 2026 at Yashobhoomi, New Delhi.
- The theme of SEMICON India 2026 is 'Silicon to Systems: Building the Ecosystem'.
- SEMICON India is organised by the India Semiconductor Mission, the Ministry of Electronics and Information Technology and SEMI.
- The Semicon India Programme was launched in 2021.
- The objective of the programme is to establish a complete semiconductor ecosystem in India.
- The programme focuses on chip design, fabrication, advanced packaging, talent development, and research and innovation.

Mind-map outline

SEMICON India and Programme 1.0

- SEMICON India conference
 - Annual flagship conference and exhibition
 - First edition: Bengaluru, 2022

- 2026 is the fifth edition
- Programme (2021)
 - Objective: complete ecosystem in India
 - Focus: design, fabrication, packaging, talent
- Organisers
 - India Semiconductor Mission
 - Ministry of Electronics and Information Technology
 - SEMI

Semicon India Programme 2.0 and Its Pillars

The Semicon India Programme 2.0 was approved in July 2026 as part of the India Semiconductor Mission, with an enhanced outlay of Rs 1,27,500 crore. It is a comprehensive, macro-level national policy framework that gives sustained, long-term financial and infrastructure support to the sector. Its six key pillars cover chip design, equipment and materials, fabrication facilities, advanced packaging, research and development, and talent development. So far 12 projects have been approved across 6 States.

- Semicon India Programme 2.0 was approved in July 2026 and is part of the India Semiconductor Mission.
- The enhanced outlay of Semicon India Programme 2.0 is Rs 1,27,500 crore.
- Programme 2.0 is a comprehensive, macro-level national policy framework.
- Programme 2.0 provides sustained, long-term financial and infrastructure support to the semiconductor sector.
- The six key pillars are chip design, semiconductor equipment and materials, fabrication facilities (FABs), advanced packaging, research and development (R&D), and talent development.
- 12 semiconductor projects have been approved across 6 States.

Mind-map outline

Programme 2.0 (approved July 2026)

- Outlay
 - Enhanced outlay: Rs 1,27,500 crore
- Six key pillars
 - Chip design
 - Equipment and materials
 - Fabrication facilities (FABs)
 - Advanced packaging
 - Research and development
 - Talent development
- Progress
 - 12 projects approved
 - Spread across 6 States
- Nature
 - Comprehensive, macro-level policy framework
 - Sustained, long-term support

Schemes, Missions and Global Partnerships

The India Semiconductor Mission (2021) is the nodal agency for semiconductor and display ecosystem schemes, and its 2.0 phase aims to make India a reliable and competitive player in the global value chain. Incentive schemes such as the PLI scheme (2020) and the Design-Linked Incentive scheme support manufacturing and design, while the Chips to Startup programme and the ChipIN Centre at C-DAC build design capacity. Internationally, India signed the iCET with the United States in 2022 and joined the Pax Silica coalition to strengthen supply-chain resilience.

- The India Semiconductor Mission (2021) is the nodal agency for implementing Semiconductor and Display Ecosystem Schemes.
- India Semiconductor Mission 2.0 aims to position India as a reliable and competitive player in the global semiconductor value chain.
- A budgetary provision of Rs 21,000 crore has been made for FY 2026-27 to support ISM 2.0.
- The Production Linked Incentive Scheme (2020) offers incentives based on incremental sales.

- The PLI scheme includes 4-6% domestic production-linked incentives for electronics companies making mobile phones and specified components.
- The Design-Linked Incentive (DLI) Scheme provides financial and infrastructure support to semiconductor design companies for over 5 years.
- The Chips to Startup (C2S) Programme gives academic institutions access to Electronic Design Automation tools, training facilities and design infrastructure.
- The ChipIN Centre at C-DAC was established under the C2S Programme and the DLI Scheme.
- India signed the India-US iCET deal in 2022 to elevate the Indo-US strategic technology partnership.
- India has joined the Pax Silica coalition to strengthen global semiconductor supply-chain resilience.

Mnemonic - Pax Silica's Silicon Stack: Minds Should Accept Infrastructure

M = Critical minerals; S = Semiconductor manufacturing; A = Advanced AI systems; I = Associated infrastructure

Mind-map outline

Schemes, missions and partnerships

- Domestic
 - ISM (2021): nodal agency
 - ISM 2.0: Rs 21,000 crore for FY 2026-27
 - PLI (2020): incremental sales
 - DLI: support over 5 years
 - C2S: EDA tools for academic institutions
- International
 - India-US iCET (2022)
 - Pax Silica: Silicon Stack

Economics

NPCI Introduces MDR on UPI Transactions

MDR framework and rates

NPCI has introduced a Merchant Discount Rate (MDR) framework for eligible person-to-merchant (P2M) UPI transactions, effective from 15 October 2026. The main rate is 0.4 per cent of value for the mid-value slab. The zero-MDR policy introduced in 2020 had exempted merchants from paying any fee for UPI and RuPay transactions.

- Framework covers eligible P2M UPI transactions above ₹2,000
- New framework comes into effect from 15 October 2026
- MDR of 0.40% of value applies to the mid-value slab
- High-value transactions of ₹75,000 and above are capped at ₹300 per transaction
- Zero-MDR policy for UPI and RuPay was introduced in 2020
- Zero-MDR policy aimed to encourage digital adoption among small businesses and informal sectors
- MDR is a fee for processing a digital payment, borne by the merchant

Mind-map outline

MDR framework and rates

- Effective 15 October 2026
- Scope: eligible P2M UPI above ₹2,000
- Mid-value slab: 0.4% of value
- High-value cap: ₹300 at ₹75,000+
- Zero-MDR policy since 2020 (earlier)

Exemptions, caps and special categories

Several categories of UPI payments are exempt from MDR, including person-to-person (P2P) transactions and P2M payments up to ₹2,000. Small merchants receiving up to ₹1 lakh per month through UPI QR codes under the P2PM category also remain exempt. Sector-specific payments and capital-market payments follow separate charging structures.

- P2P UPI transactions are not subject to MDR
- P2M UPI payments up to ₹2,000 are exempt, including through RuPay debit cards
- Transactions involving small vendors are exempt
- Small merchants receiving up to ₹1 lakh per month via UPI QR codes (P2PM) remain exempt

- Railways, telecom, insurance, fuel and agricultural inputs pay a flat MDR of ₹5 per transaction above ₹2,000
- Mutual funds, securities, stock brokers and stock dealers attract a reduced MDR of 0.02%
- Capital-market payments are subject to a ₹300 cap
- Lower capital-market rate is intended to encourage digital payments in formal financial markets

Mnemonic - Who pays nothing: Peers Vend QR Free

Peers = P2P UPI transactions are not subject to MDR; Vend = Transactions involving small vendors are exempt; QR = Small merchants receiving up to ₹1 lakh per month via UPI QR codes remain exempt

Mind-map outline

Exemptions and special categories

- Exempt
 - P2P UPI transactions
 - Small vendors
 - P2M up to ₹2,000 incl. RuPay debit
 - Small merchants up to ₹1 lakh/month QR
- Special charges
 - Sectors: flat ₹5 per transaction
 - Capital markets: 0.02%, ₹300 cap

Why MDR was introduced

The booklet lists several reasons for the MDR framework, centred on the financial sustainability of UPI. Revenue from the framework can fund infrastructure, cybersecurity and innovation. The framework also keeps P2P and low-value merchant payments free to protect consumers.

- UPI requires continuous investment in payment infrastructure and technology
- Growing digital payments need stronger systems to prevent fraud and cyber threats
- A revenue mechanism can support continued operation and expansion of UPI
- Revenue can support technological improvements and new payment solutions
- Framework keeps P2P payments and low-value merchant payments free
- Eligible small merchants continue to receive zero-MDR benefits
- A self-funded payment ecosystem can reduce dependence on continued public funding
- Small merchants and ordinary users need protection from additional costs

Mnemonic - Reasons for MDR: Fund Infra, Curb Fraud, Invent, Consumers

Fund Infra = Financial sustainability and infrastructure development; Curb Fraud = Cybersecurity and fraud prevention; Invent = Innovation through revenue for new payment solutions; Consumers = Consumer protection by keeping P2P and low-value merchant payments free

Mind-map outline

Reasons for introducing MDR

- Financial sustainability
- Infrastructure development
- Cybersecurity and fraud prevention
- Innovation
- Consumer protection
- Support for small businesses
- Less dependence on public funding

MDR, UPI and NPCI basics

MDR is a fee charged on digital payment transactions and borne by the merchant, and it is not a tax collected by the government. UPI is a real-time, 24/7 payment system that is accepted in 11 countries. NPCI was established in 2008 as a Section 8 company under the Companies Act 2013.

- MDR is distributed among banks, payment service providers and other relevant participants
- MDR is not a tax collected by the government
- MDR revenue remains within the payment ecosystem
- UPI is a real-time, 24/7, instant and interoperable payment system
- UPI is the world's largest retail fast payment system by transaction volume
- UPI is accepted in 11 countries, including Greece, Singapore, France and Qatar
- NPCI was established in 2008 as a Section 8 company under the Companies Act 2013

- NPCI was set up as an initiative of the RBI and the Indian Banks' Association
- NPCI's major payment systems are UPI, RuPay, IMPS and FASTag
- NPCI's headquarters is in Mumbai, Maharashtra

Mind-map outline

MDR, UPI and NPCI

- MDR
 - Fee borne by merchant
 - Not a government tax
- UPI
 - Real-time, 24/7 payments
 - Accepted in 11 countries
- NPCI
 - Established 2008
 - Section 8 company
 - HQ Mumbai

Summits, Reports, Indices

18th BRICS Summit 2026

Summit Overview and Theme

India hosted the 18th BRICS Summit in New Delhi in 2026, marking its 4th BRICS Chairship. The summit ran under the theme 'Building for Resilience, Innovation, Cooperation and Sustainability'. It ended with the unanimous adoption of the New Delhi Declaration, which stressed multilateralism and Global South representation. The year also marked 25 years since the BRIC concept emerged in 2001.

- 18th BRICS Summit held in New Delhi in 2026
- 2026 was India's 4th BRICS Chairship
- Earlier Indian chairships: 2012 (4th summit, New Delhi), 2016 (8th, Goa), 2021 (13th, virtual)
- Theme 2026: Building for Resilience, Innovation, Cooperation and Sustainability
- New Delhi Declaration adopted unanimously at the close of the summit
- Declaration stressed multilateralism, conflict prevention, equitable global governance and Global South representation
- 2026 marked 25 years since the BRIC concept emerged in 2001
- Resilience: strengthening ability to deal with economic, environmental and geopolitical challenges
- Innovation: promoting new technologies, research and digital solutions for development
- Cooperation and Sustainability: collaboration across sectors and environmentally sustainable, inclusive growth

Mind-map outline

Summit Overview and Theme

- Hosting
 - India's 4th BRICS Chairship
 - New Delhi, 2026
- Theme pillars
 - Resilience
 - Innovation
 - Cooperation
 - Sustainability
- Outcome
 - New Delhi Declaration, unanimous
 - 25 years since BRIC concept (2001)

Finance, Trade and Customs

The summit produced outcomes on payments, risk, MSME finance, taxation and customs. Its payment initiative aims to settle trade in local currencies and reduce exposure to exchange-rate volatility. A risk lab and a consensus on MSME financing were also announced. Tax and customs cooperation included a learning lab, an Authorised Economic Operator action plan and joint enforcement against smuggling.

- BRICS Payment Task Force: facilitates trade settlements through local currencies
- Payment initiative aims to reduce exposure to exchange-rate volatility and external financial disruptions
- BRICS Risk Lab in GIFT City, Gujarat: strengthens reinsurance capabilities
- Risk Lab addresses geopolitical risks in large infrastructure projects in the Global South
- Jaipur Consensus on MSME Financing: helps MSMEs convert unpaid invoices into working capital
- BRICS-NDB Knowledge Portal launched by the New Development Bank
- BRICS Startup Innovation Fund proposed to support early-stage ventures
- BRICS Tax Cross-Learning Lab and Authorised Economic Operator Action Plan 2026 launched
- Annual Joint Customs Enforcement Operations envisaged against smuggling networks and illicit financial flows

Mnemonic - Pay Jaipur's Risks: Pay Jaipur's Risks (P-J-R)

P = Payment Task Force: facilitates trade settlements through local currencies; J = Jaipur Consensus on MSME Financing: helps MSMEs convert unpaid invoices into working capital; R = Risk Lab: strengthens reinsurance capabilities

Mind-map outline

Finance, Trade and Customs

- Payments
 - Payment Task Force: local currencies
 - Aim: lower exchange-rate exposure
- Risk and MSME finance
 - Risk Lab: GIFT City, Gujarat
 - Jaipur Consensus: MSME financing
 - Unpaid invoices to working capital
- Development bank and funds
 - NDB Knowledge Portal
 - Startup Innovation Fund (proposed)
- Tax and customs
 - Tax Cross-Learning Lab
 - AEO Action Plan 2026
 - Annual joint customs enforcement

Science, Technology and Digital Infrastructure

The science and digital outcomes cover space, satellite data, digital public infrastructure, research infrastructure, smart grids and entrepreneurship. Terms were drafted for a proposed space council, and a repository for digital public infrastructure was approved. An action plan was adopted for a network that pools resources for high-cost megascience projects. An incubator network was launched to support innovation-led growth.

- Terms drafted for a proposed BRICS Space Council
- Cooperation under the BRICS Remote Sensing Satellite Constellation was expanded
- Collective satellite data to support disaster risk reduction and climate monitoring
- BRICS DPI Repository approved for creation
- Pilot projects started for sharing open-source digital public goods, including digital identity and payment systems
- GRAIN (Global Research Advanced Infrastructure Network): action plan adopted
- GRAIN aims to pool financial and intellectual resources for high-cost megascience projects
- Digital Centre of Excellence for Smart Grids established for AI-enabled integration of renewable energy
- BRICS Incubator Network launched to promote entrepreneurship and innovation-led growth

Mind-map outline

Science, Technology and Digital Infrastructure

- Space
 - Space Council: terms drafted
 - Remote Sensing Satellite Constellation expanded
- Digital public infrastructure
 - BRICS DPI Repository approved
 - Open-source digital public goods pilots
- Research and energy
 - GRAIN: high-cost megascience projects
 - Smart Grids Centre of Excellence
- Entrepreneurship
 - BRICS Incubator Network

Agriculture, Health, Transport and Climate

This part covers farming, health, transport, logistics and climate outcomes. Members set a target for launching a BRICS Grain Exchange and created networks on seeds, inputs and digital agriculture. Health outcomes include a healthy lifestyle roadmap and a mental well-being training network. Climate-linked items include sustainable aviation fuels and climate-resilient urban infrastructure principles.

- BRICS Grain Exchange: launch targeted by 2027
- Grain Exchange proposed in 2023 and championed by Russia
- BRICS AGRIN (Agro-Inputs, Genetic Resources and Information Network) focuses on seeds, germplasm and agricultural inputs
- Centres of Excellence on Agroecology and Regenerative Agriculture established
- BRICS Network on Digital Agriculture covers agroforestry, fisheries and aquaculture
- Joint Roadmap for Healthy Lifestyle Promotion (2026-2029) endorsed
- Training Hub Network for mental well-being led by India's NIMHANS
- BRICS BioDesign Centre spearheaded by Andhra Pradesh MedTech Zone
- Global Value Chains Action Plan 2026-2030 provides for a Technical Council and Joint Study
- Sustainable Aviation Fuels Forum launched

Mind-map outline

Agriculture, Health, Transport and Climate

- Agriculture
 - Grain Exchange by 2027
 - AGRIN: seeds, germplasm, inputs
 - Digital agriculture network
- Health
 - Healthy Lifestyle roadmap 2026-2029
 - NIMHANS: mental well-being hub
 - BioDesign Centre
- Transport and climate
 - Global Value Chains Plan 2026-2030
 - Sustainable Aviation Fuels Forum
 - Climate-resilient urban infrastructure principles

Security, Social Cooperation and Global Governance

Security outcomes include a condemnation of cross-border terrorism and an expert network for tracing fugitive offenders. BRICS CONNECT addresses capacity building and the rights of gig workers. On global governance, the summit endorsed greater African representation in the UN Security Council, and India proposed a 3R framework. China was confirmed as host of the 19th BRICS Summit in 2027.

- BRICS condemned cross-border terrorism and the 2025 Pahalgam terror attack
- BRICS urged progress towards the UN Comprehensive Convention on International Terrorism
- Expert network and repository established for tracing fugitive offenders
- BRICS CONNECT: capacity building, employability and new skills
- BRICS CONNECT aims to strengthen gig workers' rights and cross-border portability of social security

- BRICS endorsed the Ezulwini Consensus and Sirte Declaration on African representation in the UN Security Council
- India's 3R framework: Representation, Responsiveness, Rule-making
- India's 10-point roadmap for global governance reform
- BRICS endorsed WTO accession bids of Ethiopia and Iran
- China confirmed as host of the 19th BRICS Summit in 2027

Mnemonic - Reps Respond to Rules: Reps Respond to Rules

Reps = Representation: one of the three Rs in India's 3R framework for global governance reform;

Respond = Responsiveness: second of the three Rs in India's 3R framework; Rules = Rule-making: third of the three Rs in India's 3R framework

Mind-map outline

Security, Social Cooperation and Governance

- Security
 - Condemned cross-border terrorism
 - UN convention progress urged
- Crime and recovery
 - Fugitive offenders repository
 - Tracing illicit financial flows
- Labour and social security
 - BRICS CONNECT
 - Gig workers' rights
 - Cross-border social security portability
- Global governance
 - UN Security Council reform
 - 3R framework
 - WTO accession: Ethiopia, Iran

26th SCO Summit 2026

Origins and evolution

The SCO grew out of the Shanghai Five, created in 1996 by China, Russia and three Central Asian states to address border issues and build military confidence. Uzbekistan joined in 2001, leading to the SCO's establishment on 15 June 2001 in Shanghai. The SCO Charter was signed in 2002 and entered into force in 2003. Membership later widened, with India and Pakistan becoming full members in 2017, Iran in 2023 and Belarus in 2024.

- 1996: Shanghai Five created by China, Russia, Kazakhstan, Kyrgyzstan and Tajikistan
- Shanghai Five purpose: address border issues and promote military confidence-building
- 2001: Uzbekistan joined, leading to SCO establishment
- SCO established on 15 June 2001 in Shanghai
- 2002: SCO Charter signed
- 2003: SCO Charter entered into force
- 2017: India and Pakistan became full members
- 2005: India became an SCO observer
- 2023: Iran joined as a full member
- 2024: Belarus became a full member

Mind-map outline

Origins and evolution

- Shanghai Five (1996)
 - China, Russia, Kazakhstan, Kyrgyzstan, Tajikistan
 - Purpose: borders and military trust
- Uzbekistan joins (2001)
 - SCO set up in Shanghai, 15 June 2001
- Charter (2002-2003)
 - Signed 2002
 - In force 2003
- Later full members
 - India and Pakistan: 2017

- Iran: 2023
- Belarus: 2024

Institutions and membership

The SCO is a Permanent Eurasian Inter-Governmental Organisation that links Central Asia, South Asia, China, Russia and Iran. It has 10 member states and 2 observers, with headquarters in Beijing. Its Council of Heads of State is the highest decision-making body, while the Council of Heads of Government handles economic and developmental cooperation. Counter-terrorism cooperation is run by the Regional Anti-Terrorist Structure based in Tashkent.

- Described as a Permanent Eurasian Inter-Governmental Organisation
- Members: China, Russia, India, Pakistan, Kazakhstan, Kyrgyzstan, Tajikistan, Uzbekistan, Iran, Belarus
- Observers: Afghanistan and Mongolia
- Headquarters: Beijing (China)
- Council of Heads of State: highest decision-making body
- Council of Heads of Government: economic and developmental cooperation
- SCO Secretariat: permanent administrative body in Beijing
- Regional Anti-Terrorist Structure (RATS): counter-terrorism cooperation, Tashkent (Uzbekistan)
- Focus areas: regional security, counter-terrorism, economic and connectivity cooperation
- Also covers political dialogue and cultural and people-to-people exchanges

Mnemonic - Institutions by job: Decide, Develop, Administer

Decide = Council of Heads of State, the highest decision-making body; Develop = Council of Heads of Government, for economic and developmental cooperation; Administer = SCO Secretariat, the permanent administrative body in Beijing

Mind-map outline

Institutions and membership

- Institutions
 - Heads of State: highest decision-making
 - Heads of Government: economic cooperation
 - Secretariat: administration, Beijing
 - RATS: counter-terrorism, Tashkent
- Membership
 - 10 member states
 - Observers: Afghanistan, Mongolia
 - Headquarters: Beijing
- Focus areas
 - Regional security and counter-terrorism
 - Economic and connectivity cooperation
 - Political dialogue, people-to-people exchanges

Key outcomes of the 26th Summit

The 26th SCO Summit was held from 31 August to 1 September 2026 in Bishkek, Kyrgyzstan, marking the organisation's 25th anniversary under the chairmanship of President Sadyr Japarov. Leaders adopted the Bishkek Declaration as the summit's principal political document and approved 28 outcome documents. Kyrgyzstan transferred the SCO Presidency to Pakistan for 2026-27, and English was added as an official and working language.

- Dates: 31 August to 1 September 2026
- Venue: Bishkek, Kyrgyz Republic
- Chaired by President Sadyr Japarov of Kyrgyzstan
- Theme: '25 Years of the SCO: Together Towards Sustainable Peace, Development and Prosperity'
- Bishkek Declaration adopted as the summit's principal political document
- Leaders approved 28 outcome documents
- English added as an official and working language, alongside Russian and Chinese
- Presidency transferred from Kyrgyzstan to Pakistan for 2026-27
- Lahore designated SCO Tourism and Cultural Capital for 2026-27

- Pakistan expected to host the 2027 Council of Heads of State meeting

Mind-map outline

26th Summit outcomes

- Venue and dates
 - Bishkek, Kyrgyz Republic
 - 31 August to 1 September 2026
- Documents
 - Bishkek Declaration: principal document
 - 28 outcome documents
- Language and presidency
 - English added as official language
 - Presidency to Pakistan for 2026-27
- Outreach and youth
 - African Union Commission MoU
 - 1st SCO Civilisation Dialogue Forum, India, 2026

India and the SCO

India frames its SCO engagement around three pillars: Security, Connectivity and Opportunity for regional cooperation. The SCO gives India institutional access to Central Asia, bypassing the lack of direct overland routes, and supports long-term energy and resource security. Frictions include the China-Pakistan strategic axis, India's refusal of the BRI because CPEC passes through disputed territories, and slow progress on payment arrangements and transport corridors. India also took part in the 6th World Nomad Games 2026 with its largest-ever contingent of 87 members.

- India's SCO vision rests on three pillars: Security, Connectivity and Opportunity
- SCO provides institutional access to Kazakhstan, Kyrgyzstan, Tajikistan and Uzbekistan
- Chabahar Port and INSTC reduce dependence on Pakistan-controlled transit
- RATS treats terror financing and radicalisation as collective regional concerns
- Energy engagement secures long-term access to uranium, minerals, oil and natural gas
- Soft power pillars include Ayurveda and Buddhism
- India refuses BRI because CPEC passes through disputed territories
- Implementation gap: slow progress in payment arrangements and transport corridors
- PM Narendra Modi attended the opening of the 6th World Nomad Games in Bishkek
- India fielded its largest-ever contingent of 87 members at the 2026 Games

Mnemonic - Three pillars of India's SCO vision: SCO: Security, Connectivity, Opportunity
S = Security; C = Connectivity; O = Opportunity for regional cooperation

Mind-map outline

India and the SCO

- Three pillars
 - Security
 - Connectivity
 - Opportunity
- Gains
 - Access to landlocked Central Asia
 - Chabahar Port and INSTC
 - Energy: uranium, minerals, oil, gas
- Frictions
 - China-Pakistan strategic axis
 - Refuses BRI over CPEC
 - Slow payment and transport progress
- World Nomad Games 2026
 - PM Modi at opening in Bishkek
 - Largest-ever contingent: 87 members

81st UN General Assembly, P4M and Arctic Affairs

P4M Summit at the 81st UNGA

India's External Affairs Minister co-chaired the inaugural Partners for Multilateralism, International Law, Peace and Prosperity (P4M) Summit with European Council President Antonio Costa, on the sidelines of the 81st UNGA in New York. The P4M framework is built upon the Carney Doctrine on middle-power solidarity. It is designed as an open and flexible framework rather than an exclusive bloc, and it seeks to strengthen international law, peace, multilateral cooperation and inclusive prosperity.

- 81st UNGA opened on 8 September 2026 in New York
- UNGA theme: Restoring Trust, Managing Transformation: A United Nations That Delivers for All
- P4M co-chaired by India's External Affairs Minister and European Council President Antonio Costa
- P4M built upon Canadian PM Mark Carney's Carney Doctrine on middle-power solidarity, delivered at Davos
- Conceived in response to erosion of global consensus, trade weaponisation and escalating conflicts
- Conceptualised by leaders of the EU, Brazil, Kenya and Canada
- Co-sponsored by India, Australia and Barbados
- Designed as an open and flexible framework, not an exclusive bloc or permanent bureaucracy

Mind-map outline

P4M Summit

- Origin
 - Carney Doctrine, delivered at Davos
 - Response to trade weaponisation and conflicts
- Backers
 - Conceptualised: EU, Brazil, Kenya, Canada
 - Co-sponsors: India, Australia, Barbados
- Purpose
 - Strengthen international law and peace
 - Reform multilateral institutions

India's Position on Reformed Multilateralism

India argued that disruptions in fuel, food, fertiliser and finance hit developing countries and the Global South disproportionately. It linked supply-chain risks to a call for de-risking and diversification. India also called for greater representation of developing countries, more inclusive decision-making, transparency and comprehensive reform of the UN Security Council, and for guardrails on emerging technologies.

- 4F crisis: disruptions involving Fuel, Food, Fertiliser and Finance
- 4F disruptions have a disproportionate impact on developing countries and the Global South
- Supply-chain risks: disruptions, strategic chokepoints and excessive economic dependencies
- Called for de-risking and diversification to build more resilient global supply chains
- Geopolitical fragmentation and strategic competition put existing multilateral institutions under pressure
- Called for greater representation of developing countries and greater transparency in global institutions
- Pursues UNSC reform through the G4 and L.69 groupings
- Called for international frameworks and guardrails for AI, space security and digital technologies

Mnemonic - 4F crisis: Fire Feeds Farms and Funds

Fire = Fuel; Feeds = Food; Farms = Fertiliser; Funds = Finance

Mind-map outline

India's reformed multilateralism

- 4F crisis
 - Fuel, Food, Fertiliser, Finance
 - Hits developing countries disproportionately
- Supply chains
 - Risks: chokepoints, dependencies
 - Remedy: de-risking and diversification
- Global institutions
 - Greater representation of developing countries

- Comprehensive UNSC reform
- Emerging technologies
- AI, space security, digital technologies

Visegrad Group (V4)

The Visegrad Group is a regional cooperation grouping of four Central European countries, established in 1991 after the end of the Cold War and the collapse of the Soviet bloc. All four members belong to both the European Union and NATO. At the V4 plus India meeting, the V4 unanimously supported India's candidature for permanent membership of the UN Security Council.

- Named after Visegrad, a town in Hungary
- Established in 1991 after the end of the Cold War and the collapse of the Soviet bloc
- Members: Czech Republic, Hungary, Poland and Slovakia
- All four members are members of both the EU and NATO
- Cooperation covers political and diplomatic coordination, defence and security
- Cooperation also covers economic cooperation, energy and culture
- Unanimously supported India's candidature for permanent membership of the UN Security Council

Mnemonic - V4 members: Czechs Have Polish Sausages

C = Czech Republic; H = Hungary; P = Poland; S = Slovakia

Mind-map outline

Visegrad Group (V4)

- Origin
 - Named after a town in Hungary
 - Founded 1991, after Cold War
- Members
 - Czech Republic
 - Hungary
 - Poland
 - Slovakia
- Memberships
 - All four in EU and NATO
- India link
 - Unanimous support for India's UNSC candidature

US-Denmark-Greenland Agreement and Greenland's Strategic Role

A trilateral security agreement among the United States, Denmark and Greenland was signed on the sidelines of the 81st UNGA. It builds on the 1951 US-Denmark Defence Agreement, expands US activities in Greenland and restricts military presence by non-NATO states through joint approval. It does not transfer sovereignty over Greenland and recognises Greenland's right to self-determination.

- Signed by the United States, Denmark and Greenland on the sidelines of the 81st UNGA
- Builds on the 1951 US-Denmark Defence Agreement while reaffirming Greenland's right to self-determination
- Expands activities at Pituffik Space Base, formerly Thule Air Base
- US military access to Narsarsuaq (southern Greenland) and Mestersvig (eastern Greenland)
- Restricts military presence by non-NATO states through joint approval
- Does not transfer sovereignty over Greenland to the United States
- Greenland lies within the GIUK gap, a key chokepoint for monitoring North Atlantic naval movements
- Large untapped rare earth reserves; Transpolar Sea Route opening as Arctic ice melts

Mnemonic - GIUK gap: Gentle Ice Upsets Kayakers

G = Greenland; I = Iceland; U = United (part of United Kingdom); K = Kingdom (part of United Kingdom)

Mind-map outline

US-Denmark-Greenland agreement

- Agreement terms
 - Builds on 1951 defence agreement

- Pituffik, Narsarsuaq, Mestersvig access
- Non-NATO presence needs joint approval
- No transfer of sovereignty
- Strategic value
 - Between North America and Europe
 - GIUK gap chokepoint
 - Missile early-warning infrastructure
- Resources and routes
 - Rare earth reserves
 - Transpolar Sea Route opening

Arctic Council

The Arctic Council is an inter-governmental platform established in 1996 through the Ottawa Declaration. It provides a forum for discussion and cooperation on the Arctic environment and sustainable development, with its main focus on climate change and its impact on the Arctic region. Eight Arctic nations are members, and India has been an observer since 2013.

- Type: inter-governmental platform
- Established in 1996 through the Ottawa Declaration
- Purpose: forum for cooperation on the Arctic environment and sustainable development
- Main focus: climate change and its impact on the Arctic region
- Members: Canada, Denmark, Finland, Iceland, Norway, Russia, Sweden and the United States
- Chairmanship alternates every 2 years; current chair is Denmark (2025-2027)
- India has been an observer since 2013
- India's first Arctic research station, Himadri (2008), is at Ny-Alesund, Svalbard, Norway

Mind-map outline

Arctic Council

- Basics
 - Inter-governmental platform
 - Established 1996, Ottawa Declaration
- Members
 - 8 Arctic nations
 - India observer since 2013
- Leadership
 - Chair rotates every 2 years
 - Denmark chairs 2025-2027
- Focus
 - Climate change and Arctic region

Global Gender Gap Index 2026

Report Basics and Scoring

The Global Gender Gap Report is an annual flagship publication of the World Economic Forum, introduced in 2006. It tracks how far countries have closed gender disparities rather than only counting women's absolute outcomes. Each country is scored from 0 to 1, and the scores are also expressed on a 0-100 scale where 100 means complete parity. The index is built on four dimensions.

- Introduced in 2006 by the World Economic Forum
- The 2026 report is the 20th edition
- Annual flagship publication of the World Economic Forum
- Tracks closing of gender disparities, not women's absolute outcomes
- Four dimensions: Economic Participation and Opportunity, Educational Attainment, Health and Survival, Political Empowerment
- Score of 0 reflects absolute inequality; score of 1 means complete gender equality
- Scores expressed on a 0-100 scale; 100 represents complete parity
- World Economic Forum is a Switzerland-based think tank and advocacy group
- The Forum was founded by Klaus Schwab in 1971

Mind-map outline

Report Basics and Scoring

- Who publishes it
 - World Economic Forum
 - Annual flagship publication
- What it measures
 - Closing of gender disparities
 - Four dimensions
- Scoring
 - Score from 0 to 1
 - 0-100 scale, 100 is parity

Global Highlights and Leadership

Iceland retained the first position for the 17th consecutive year, while Chad ranked last among 145 countries. The world has closed 69.2% of the overall gender gap, improving by only 0.4 percentage points in the year. At the current pace, complete parity could take about 120 years. Women hold 19.1% of CEO positions globally.

- Iceland retained 1st position for the 17th consecutive year
- Iceland closed around 93% of its gender gap
- Finland and Norway followed Iceland
- Australia entered the global top 10 for the first time
- Chad ranked 145th, lowest among the 145 countries covered
- Pakistan and Iran were among the three lowest-ranked countries
- World has closed 69.2% of the overall gender gap
- Global parity improved by only 0.4 percentage points over the previous year
- Complete global parity could take about 120 years at the present rate of progress
- Women hold 19.1% of CEO positions globally

Mnemonic - Progress Snapshot: Sixty-Nine Done, One-Twenty to Go

Sixty-Nine = World has closed 69.2% of the overall gender gap; One-Twenty = Complete global parity could take about 120 years at present progress

Mind-map outline

Global Highlights and Leadership

- Leaders
 - Iceland: 17th year at top
 - Finland and Norway follow
 - Australia enters top 10 first time
- Bottom
 - Chad ranked 145th
 - Pakistan and Iran also lowest
- Pace of progress
 - World closed 69.2% of gap
 - Parity improved only 0.4 points
 - Complete parity in about 120 years
- Women's leadership
 - 19.1% of CEO positions
 - Career breaks: nearly twice as likely
 - Fewer than 1 in 5 AI engineers

India's Performance

India retained its 131st position among 145 countries, with an overall gender parity score of 64.5%. Its dimension scores are 96.6% in educational attainment, 95.6% in health and survival, 41.2% in economic participation and opportunity, and 24.5% in political empowerment. Women hold only 5.9% of ministers and 13.1% of senior leadership and managerial positions. The booklet notes that progress in education and health has not yet carried over to the economic and political dimensions.

- India's rank: 131st of 145 countries, retaining its position
- Overall gender parity score 64.5%, against a global average of 69.2%
- Educational attainment: 96.6% parity

- Health and survival: 95.6% parity
- Economic participation and opportunity: 41.2%
- Political empowerment: 24.5%
- Women are 5.9% of ministers, compared with a peak of 30% in 2019
- Women are 13.6% (74) of the 18th Lok Sabha in 2024, down from 14.4% in 2019
- Women occupy 13.1% of senior leadership and managerial positions
- Only 17% of CEOs and 20% of board members in Indian corporates are women

Mind-map outline

India's Performance

- Overall
 - Rank 131st of 145
 - Score 64.5% vs global 69.2%
- Dimension scores
 - Education 96.6%
 - Health and Survival 95.6%
 - Economic 41.2%
 - Political 24.5%
- Representation
 - Ministers 5.9%
 - Lok Sabha women 13.6% (2024)
 - Senior leadership 13.1%

Barriers and Government Initiatives

Women spend nearly five hours a day on unpaid household chores, three times more than men, and this unpaid work is valued at 22.7 lakh crore or 7.5% of GDP. Labour market, workplace safety and childcare issues continue to hinder women's participation. The Nari Shakti Vandan Adhiniyam mandates 33% representation for women in legislatures. Schemes such as Beti Bachao Beti Padhao, PM Matru Vandana Yojana and Pradhan Mantri Jan-Dhan Yojana support girls' education, maternal health and financial access.

- Women spend nearly 5 hours daily on unpaid household chores, three times more than men
- Unpaid work is valued at 22.7 lakh crore, or 7.5% of GDP
- Most women are in informal, low-paid roles despite rising participation
- Workplace safety concerns and inadequate childcare are major hurdles
- 40% of girls aged 15-18 are not in school
- Female labour participation grew from 23.3% (2017-18) to 41.7% (2023-24)
- Nari Shakti Vandan Adhiniyam (106th Constitutional Amendment, 2023) mandates 33% representation for women in legislatures
- Beti Bachao Beti Padhao (2015) targets the falling child sex ratio
- Over 28 crore women have Jan Dhan bank accounts
- Maternal health programmes have cut the maternal mortality rate from 174 to 97

Mind-map outline

Barriers and Government Initiatives

- Barriers
 - Unpaid work nearly 5 hours daily
 - Most women in informal low-paid roles
 - 40% of girls aged 15-18 not in school
- Initiatives
 - Nari Shakti Vandan Adhiniyam (2023)
 - Beti Bachao Beti Padhao (2015)
 - PM Matru Vandana Yojana (2017)
 - Jan Dhan: over 28 crore women accounts

Environment & Biodiversity

Swachh Vayu Sarvekshan 2026

Swachh Vayu Diwas and Clean Air Day

Swachh Vayu Diwas 2026 was observed on 7 September 2026 and coincides with the International Day of Clean Air for Blue Skies. The International Day was designated by the UN General Assembly in 2019 and carries the 2026 theme 'Clean Air for Climate Action'. The occasion marked the presentation of the 5th Swachh Vayu Sarvekshan Awards.

- Swachh Vayu Diwas 2026 was observed on 7 September 2026.
- It coincides with the International Day of Clean Air for Blue Skies.
- The UN General Assembly designated the International Day in 2019.
- The 2026 theme of the International Day is 'Clean Air for Climate Action'.
- India observes the occasion as Swachh Vayu Diwas.
- The occasion marked presentation of the 5th Swachh Vayu Sarvekshan Awards.

Mind-map outline

Swachh Vayu Diwas 2026

- Date: 7 September 2026
- Coincides with Clean Air Day
- Clean Air Day designated by UNGA in 2019
- Theme: Clean Air for Climate Action
- 5th Sarvekshan Awards presented

National Clean Air Programme and NCAP targets

The National Clean Air Programme was launched by MoEFCC in 2019 as a national-level strategy covering 130 non-attainment cities along with million-plus cities. Its first target was a 20-30 per cent cut in PM10 and PM2.5 by 2024 using 2017 as the baseline, later enhanced to a 40 per cent cut in particulate matter by 2026. The revised target has not been fully achieved, as only 26 of 130 cities recorded a reduction of 40 per cent or more.

- NCAP was launched by MoEFCC in 2019 as a national-level strategy.
- NCAP covers 130 non-attainment cities along with million-plus cities.
- Non-attainment cities are those that failed to comply with NAAQS over a specified assessment period.
- Initial NCAP target: 20-30% reduction in PM10 and PM2.5 by 2024, with 2017 as baseline year.
- Target was enhanced to a 40% reduction in particulate matter concentration by 2026.
- Only 26 of 130 cities recorded a reduction of 40% or more.
- In 2025-26, 108 cities recorded reductions in annual PM10 compared to 2017-18.
- 72 cities achieved reductions of more than 20% in FY 2025-26.
- 14 cities met the prescribed NAAQS.

Mnemonic - NCAP target years: Seventeen base, twenty-four start, forty by twenty-six
Seventeen = 2017 is the baseline year; Twenty-four = 2024 is the initial target year; Forty = 40% enhanced reduction target; Twenty-six = 2026 is the enhanced target year

Mind-map outline

National Clean Air Programme

- Launched by MoEFCC, 2019
- Covers 130 non-attainment cities
- Targets
 - 20-30% cut by 2024 (2017 base)
 - 40% particulate cut by 2026
- Progress 2025-26
 - 108 cities cut PM10 vs 2017-18
 - 72 cities cut over 20%
 - 26 cities cut 40% or more
 - 14 cities met NAAQS

Swachh Vayu Sarvekshan and city rankings

The Swachh Vayu Sarvekshan was launched in 2022-23 as an annual assessment of the 130 NCAP cities. Cities are grouped into three population categories and evaluated across five key parameters. The 2026 edition introduced ward-level rankings for the first time, and the city-level toppers in each category are listed in the booklet.

- Swachh Vayu Sarvekshan was launched in 2022-23 as an annual assessment of 130 NCAP cities.

- Cities are grouped as Category I (above 10 lakh), Category II (3-10 lakh) and Category III (below 3 lakh).
- Cities are evaluated on Ambient Air Quality, Source Apportionment, Action Plan Implementation, Institutional Capacity and Public Participation.
- Category I (2026): Lucknow 1st (198), Indore 2nd (197), Jabalpur 3rd (195).
- Category I bottom 5: Chennai, Jamshedpur, Kota, Kolkata and Madurai.
- Delhi ranked 30th in Category I.
- Category II (2026): Rourkela 1st (198.5); Firozabad and Guntur joint 2nd (195); Amravati 3rd (194).
- Category III (2026): Kalinganagar 1st (198.5); Angul 2nd (196.5); Talcher 3rd (193.5).
- The 2026 edition introduced Ward-level Rankings for the first time.

Mnemonic - Five key parameters: A Sensible Approach In Public

A = Ambient air quality; S = Source apportionment; A = Action plan implementation; I = Institutional capacity; P = Public participation

Mind-map outline

Swachh Vayu Sarvekshan

- Three population categories
 - Category I: above 10 lakh
 - Category II: 3-10 lakh
 - Category III: below 3 lakh
- Five key parameters
 - Ambient air quality
 - Source apportionment
 - Action plan implementation
 - Institutional capacity
 - Public participation
- Ward-level rankings (2026)

Pollution control tools, measures and laws

The booklet lists monitoring tools, field measures, emergency action and legal provisions against air pollution. Monitoring includes the National Air Quality Index and SAFAR, while field measures include the Turbo Happy Seeder and Pusa Bio-Decomposer. The legal base includes the Air Act 1981, Water Act 1974 and Environment Protection Act 1986, along with Article 51A(g) and Article 21.

- National Air Quality Index (2014) is a tool for communicating air quality status in easy terms.
- Turbo Happy Seeder can uproot stubble and also sow seeds in the area cleared.
- Pusa Bio-Decomposer turns crop residue into manure in 15-20 days and was developed by IARI.
- Graded Response Action Plan (2017) is a set of emergency measures.
- GRAP measures kick in only when air quality shifts from Poor to Very Poor.
- SAFAR is a Ministry of Earth Sciences initiative to monitor city-wide and localised air quality.
- Bharat Stage VI (2020) made the leap from BS-IV to BS-VI mandatory nationwide.
- Air Act 1981, Water Act 1974 and Environment Protection Act 1986 are key pollution control regulations.
- The 42nd Constitutional Amendment (1976) inserted Clause (g) in Article 51A, making protection of the natural environment a Fundamental Duty.
- Article 21 has been interpreted to include the right to a healthy environment.

Mnemonic - Three pollution laws: Always Watch the Environment

A = Air (Prevention and Control of Pollution) Act, 1981; W = Water (Prevention and Control of Pollution) Act, 1974; E = Environment Protection Act, 1986

Mind-map outline

Air pollution controls

- Monitoring tools
 - National AQI (2014), CPCB
 - SAFAR, Ministry of Earth Sciences
- Field measures
 - Turbo Happy Seeder
 - Pusa Bio-Decomposer, 15-20 days

- Emergency and vehicle norms
 - GRAP (2017) at Poor to Very Poor
 - BS-VI mandatory from 2020
- Legal framework
 - Air Act 1981
 - Water Act 1974
 - EP Act 1986
 - Article 51A(g) and Article 21

Sports, S&T & Defence

EOS-05: India's 1st Dedicated GEO Satellite

EOS-05 (GISAT-1A) and the GSLV-F17 launch

ISRO launched Earth Observation Satellite-05 (EOS-05), also called GISAT-1A, aboard GSLV-F17 from the Satish Dhawan Space Centre, Sriharikota. The booklet calls it India's first dedicated imaging satellite, designed to operate from a sub-geosynchronous orbit about 35,786 km above Earth. The launch was the 19th operational flight of GSLV and was described as validating India's cryogenic upper-stage technology for future human spaceflight.

- Launch vehicle: GSLV-F17, marked as the 19th operational flight of GSLV
- Launch site: Satish Dhawan Space Centre, Sriharikota
- EOS-05 is India's first dedicated imaging satellite for sub-geosynchronous operations
- Mass of EOS-05: approx 2,367 kg
- Heaviest payload successfully injected into orbit by a GSLV
- GSLV three stages: solid core first stage with 4 liquid strap-on boosters
- GSLV second stage: liquid propellant
- GSLV third stage: cryogenic
- Nickname of EOS-05: 'Eye in the Sky'
- EOS-05 carries optical and infrared imaging payloads

Mnemonic - GSLV stage order: Solid Lamps Cool

S = Solid core first stage with 4 liquid strap-on boosters; L = Liquid propellant second stage; C = Cryogenic third stage

Mind-map outline

EOS-05 launch on GSLV-F17

- Vehicle
 - Marked 19th operational GSLV flight
 - Heaviest GSLV-injected payload
- Three stages
 - Solid first stage, 4 liquid strap-ons
 - Liquid second stage
 - Cryogenic third stage
- Satellite
 - Nickname: Eye in the Sky
 - Mass approx 2,367 kg

Satellite orbits: LEO, MEO and GEO

An orbit is the curved path a satellite follows around Earth under gravity, and the choice of orbit shapes coverage, resolution, revisit time, delay and fuel need. LEO lies about 160-2,000 km up and offers low latency and high-resolution observation. MEO spans about 2,000-35,786 km and hosts navigation systems, while GEO is a special case of geosynchronous orbit at about 35,786 km.

- Orbit: curved path followed by a satellite around Earth under gravity
- Orbit choice determines coverage, image resolution, revisit time, delay, fuel and mission
- LEO altitude: approx 160-2,000 km; orbital period about 90-120 minutes
- LEO features: low communication latency, suited to large constellations
- Sun-synchronous orbit: LEO subtype giving relatively consistent solar illumination
- Very Low Earth Orbit: very high-resolution observation but greater atmospheric drag

- MEO altitude: approx 2,000 km to 35,786 km
- MEO use: navigation systems such as GPS and Europe's Galileo
- Geosynchronous orbit period equals Earth's rotation (23 hrs, 56 mins, 4 secs)
- GEO: zero-degree inclination, directly over the Equator, moving in Earth's rotation direction
- A GEO satellite appears fixed over one point on Earth

Mnemonic - Geostationary conditions: Zebras Eat Slowly

Z = Zero-degree inclination; E = Orbit lies directly over the Equator; S = Same direction as Earth's rotation

Mind-map outline

Satellite orbits

- LEO
 - 160-2,000 km altitude
 - Period about 90-120 minutes
 - Low latency; Sun-synchronous subtype
- MEO
 - 2,000-35,786 km altitude
 - Navigation: GPS, Galileo
- GEO
 - About 35,786 km altitude
 - Zero inclination over Equator
 - Appears fixed over one point

Why GEO imaging matters and its uses

A LEO satellite observes a location only when it passes overhead, which leaves intervals between observations. A geosynchronous imaging platform can give persistent, wide-area monitoring, useful for fast events such as cyclones, cloudbursts, flash floods and forest fires. The booklet lists EOS-05 applications under disaster management, strategic and national security, and agriculture and environment.

- LEO satellite sees a particular location only when it passes overhead
- Geosynchronous platform provides persistent and wide-area monitoring
- Useful for tracking cyclones, cloudbursts, flash floods and forest fires
- Frequent observations support weather monitoring, disaster management, early warning and emergency response
- Disaster management: near real-time monitoring of cyclones, floods and cloudbursts
- Strategic security: continuous surveillance of land borders and maritime domain
- Agriculture and environment: monitoring forest fires, vegetation health and crop cycles
- Optical sensors cannot penetrate thick clouds
- High altitude limits sub-meter spatial resolution
- Crop insurance claims streamlined by replacing manual experiments with objective satellite indices

Mnemonic - Applications of EOS-05: Dawn Sees All

D = Disaster management: near real-time monitoring of cyclones, floods and cloudbursts; S = Strategic and national security: surveillance of land borders and maritime domain; A = Agriculture and environment: forest fires, vegetation health and crop cycles

Mind-map outline

GEO imaging and its uses

- LEO limitation
 - Observes a spot only on overpass
 - Gaps between observations
- GEO advantage
 - Persistent wide-area monitoring
 - Fast events: cyclones, flash floods
- EOS-05 uses
 - Disaster management
 - Border and maritime surveillance
 - Forest fire and crop health

NISAR: NASA-ISRO radar satellite

NISAR (NASA-ISRO Synthetic Aperture Radar) is described as the first joint Earth-observation satellite of NASA and ISRO. As a radar satellite it uses microwave pulses to capture sharp images in darkness and through clouds, unlike cameras that rely on visible light. Its data supports study of solid Earth processes, natural hazards, biomass and crop health.

- First joint Earth-observation satellite by NASA and ISRO
- Uses microwave pulses, so it works in darkness and through clouds
- World's first radar satellite using dual S- and L-band frequencies
- Designed for a 3-year mission life
- Orbit: 747 km, Sun-Synchronous (Low Earth) Orbit with a 12-day repeat cycle
- The NASA-side radar, with a wavelength of about 24 cm, was developed by NASA's Jet Propulsion Laboratory (JPL)
- Sweep SAR captures wide 3D high-resolution images over a 242 km swath
- Large 12-metre antenna for receiving signals
- Detects minute ground shifts for tectonic and structural analysis
- Tracks biomass, deforestation, crop health and soil moisture; assesses earthquakes, floods, landslides and volcanoes

Mind-map outline

NISAR radar satellite

- Partners
 - NASA and ISRO
 - First joint EO satellite
- Radar technology
 - Microwave pulses; works through clouds
 - Dual S- and L-band frequencies
 - Sweep SAR; 242 km swath
- Orbit and life
 - 747 km, sun-synchronous
 - 12-day repeat cycle
 - 3-year designed mission life

Miscellaneous

Miscellaneous: In News (September 2026)

Maritime and Territorial Disputes

A Pakistani naval vessel, PNS Hunain, collided with the Indian Navy warship INS Kolkata in the North Arabian Sea, and India lodged a diplomatic protest after calling the conduct unsafe. The 1991 India-Pakistan agreement on advance notice of military exercises is a confidence-building measure meant to reduce accidental escalation. The Falklands dispute between Argentina and the United Kingdom has resurfaced over sovereignty and oil exploration, and the United States has moved to rename Lake Ontario, a step Canada opposed.

- PNS Hunain collided with INS Kolkata in the North Arabian Sea
- India called the Pakistani vessel's conduct unsafe and said the incident was in international waters
- India lodged a diplomatic protest with Pakistan
- Similar earlier incident: INS Godavari and PNS Babur in the Gulf of Aden, June 2011
- 1991 Agreement: Agreement on Advance Notice of Military Exercises, Manoeuvres and Troop Movements
- 1991 Agreement is a confidence-building measure to reduce misinterpretation and accidental escalation
- Article 10: ships and submarines should not approach within 3 nautical miles in international waters
- COLREGs set internationally recognised rules of navigation and collision avoidance for vessels at sea
- UK says it has exercised continuous and peaceful authority over the Falklands since 1833; Argentina disputes this
- US executive order to rename Lake Ontario 'Lake America'; Canada opposed, citing its Wendat origins

Mnemonic - 1991 Agreement distance and date: Three miles off, six April sign-off
Three miles off = Ships and submarines should not approach within 3 nautical miles in international waters; six April = The 1991 India-Pakistan Agreement was signed on 6 April 1991

Mind-map outline

Maritime and Territorial Disputes

- Naval collision, North Arabian Sea
 - PNS Hunain vs INS Kolkata
 - India: unsafe conduct, protest lodged
 - Similar case: INS Godavari vs PNS Babur (2011)
- 1991 Agreement
 - Advance notice of military exercises
 - Article 10: 3 nautical miles
 - Aim: fewer accidents between navies
- Falklands dispute
 - UK: continuous authority claimed
 - Argentina: Las Malvinas claim
 - 2013 referendum for British status
- Lake Ontario renaming
 - US executive order: Lake America
 - Canada: Wendat origin objection

International Conventions and Diplomacy

The European Commission President has proposed exploring an 'Associate Member' relationship that could let Canada take part more closely in selected EU frameworks without full membership. Under the UN Convention on Certain Conventional Weapons, 128 countries reached a consensus on a working definition of lethal autonomous weapons systems. CITES regulates international trade in endangered wild species through three appendices, and the Montreal Protocol governs the phase-out of ozone-depleting substances and the Kigali Amendment mandates the phase-down of hydrofluorocarbons.

- EU proposal to explore an 'Associate Member' relationship, illustrated with Canada
- 'Associate Membership' is not an established membership category under the EU's foundational treaties
- Treaty revision would require all 27 EU Member States to agree
- Association Agreement under Article 217 of the TFEU is a more flexible alternative
- Association Agreement needs unanimous approval in the Council and the consent of the European Parliament
- CCW adopted in 1980 and entered into force in 1983
- 128 countries reached consensus on a working definition of Lethal Autonomous Weapons Systems (LAWS)
- CITES has 185 Parties; India became a Party in 1976
- CITES species are listed in three appendices by conservation status and trade sensitivity
- Montreal Protocol (1987) led to a 99% phase-out of controlled ozone-depleting substances; Kigali Amendment (2016) mandates phase-down of HFCs

Mnemonic - CITES appendices: Eager Monkeys Need Care

E = Appendix I: endangered species, trade allowed only in rare cases; M = Appendix II: species that may become endangered without regulation; N = Appendix III: species protected nationally, needing global cooperation

Mind-map outline

International Conventions and Diplomacy

- EU Associate Membership
 - Not an established EU membership category
 - Treaty revision needs all 27 members
 - Article 217 Association Agreement
- CCW and LAWS
 - Adopted 1980, in force 1983
 - 128 countries agreed LAWS definition
- CITES

- 185 Parties; India joined 1976
- Appendix I: most endangered species
- Appendix II and III: regulated trade
- Ozone treaties
 - Vienna Convention (1985)
 - Montreal Protocol (1987): 99% phase-out
 - Kigali Amendment (2016): HFC phase-down

Governance, Standards and Schemes

The PM Vishwakarma scheme completed three years of implementation in September 2026 and supports artisans in 18 traditional trades. Registered Unrecognised Political Parties (RUPPs) are under scrutiny over political funding transparency, with the Election Commission of India registering parties under the Representation of the People Act, 1951. The Legal Metrology (Indian Standard Time) Rules, 2026 make IST the single mandatory time reference for official purposes, and Karnataka has granted Tulu additional administrative language status.

- PM Vishwakarma scheme is fully funded by the Government of India and runs from FY 2023-24 to FY 2027-28
- Covers 18 traditional trades in rural and urban areas
- Only one member per family can apply; applicant must be over 18 years of age
- Applicants who benefited from PMEGP or PM Swanidhi in the past 5 years are not eligible
- RUPPs are registered with ECI but have not acquired status of a recognised State or National Party
- Section 29A RPA 1951: registration of political parties; Section 29C: contribution reports
- Supreme Court in ADR v. Union of India (2024) struck down the Electoral Bond Scheme
- Legal Metrology (Indian Standard Time) Rules, 2026 are notified under Section 52 of the Legal Metrology Act, 2009
- IST is 5 hours 30 minutes ahead of UTC and is based on 82.5 degrees East longitude
- Tulu was granted status of additional administrative language in Karnataka; Tulu is not in the Eighth Schedule

Mind-map outline

Governance, Standards and Schemes

- PM Vishwakarma
 - Three years completed, Sep 2026
 - 18 traditional trades
 - Collateral-free enterprise loans
- Political funding
 - RUPPs registered but not recognised
 - Section 29C contribution reports
 - Electoral Bond Scheme struck down (2024)
- Indian Standard Time
 - UTC + 5:30; 82.5 degrees East
 - CSIR-NPL is the custodian
 - Legal Metrology Rules 2026
- Tulu language status
 - Additional administrative language, Karnataka
 - Tigalari: Tulu script

Economy, Energy and Urbanisation

India's foreign exchange reserves rose to a record high, moving the country to fourth place globally, with the rise linked to RBI concessional swap arrangements tied to FCNR(B) deposits. The Union Cabinet approved a scheme to promote new surface coal and lignite gasification projects, building on the National Coal Gasification Mission of 2021. The India Electric Mobility Index 2025 ranks States and UTs on EV progress, while the Kishau Multipurpose Dam and the Census Town criteria address water sharing and urbanisation.

- India's forex reserves moved to 4th position globally, ahead of Russia and after China, Japan and Switzerland

- FCNR(B) deposits are term deposits maintained by NRIs in foreign currency with authorised Indian banks
- Coal gasification scheme builds on the National Coal Gasification Mission (2021) and aims to produce syngas
- Target to gasify around 75 million tonnes of coal or lignite by 2030
- Government provides financial assistance of up to 20% of expenditure on plant and machinery
- Kishau Multipurpose Dam: inter-state project on the Tons River, a tributary of the Yamuna
- Union Government provides 90% of the water component cost; the 6 participating States/UT contribute the remaining 10%
- India Electric Mobility Index 2025 by NITI Aayog and WRI India scores States and UTs on 16 indicators
- IEMI pillars: Transport Electrification Progress 50%, Charging Infrastructure Readiness 30%, EV Research and Innovation 20%
- Census Town (1981 criteria): at least 5,000 persons, 75% of male workers in non-agricultural work, 400 persons per sq km

Mnemonic - EV Mobility Index pillar weights: Half, Thirty, Twenty: Transport, Charging, Research
 Half = Transport Electrification Progress carries 50%; Thirty = Charging Infrastructure Readiness carries 30%; Twenty = EV Research and Innovation carries 20%

Mind-map outline

Economy, Energy and Urbanisation

- Forex reserves
 - 4th position globally
 - RBI swap arrangements
 - FCNR(B) deposit inflows
- Coal gasification
 - Syngas from surface coal and lignite
 - 75 million tonnes by 2030
 - Up to 20% of plant and machinery cost
- Kishau Dam
 - Tons River, Yamuna basin
 - Centre funds 90% of water cost
 - 6 States/UT share the rest
- Census Town
 - At least 5,000 persons
 - 75% male workers non-agricultural
 - 400 persons per sq km
- EV Mobility Index 2025
 - 16 indicators, 0-100 scale
 - Transport pillar 50%
 - Delhi tops overall

Science, Environment and Culture

PRATUSH is a proposed Indian mission, while CosmoCube is a proposed lunar-orbiting mission led by scientists associated with the Royal Astronomical Society and University of Cambridge, both aimed at detecting the 21-cm signal from the early universe, while NASA's Nancy Grace Roman Space Telescope has been launched to study dark energy and exoplanets. Five young orangutans rescued in Odisha were moved to a zoo, and orangutans are native only to Borneo and Sumatra. World Rhino Day on 22 September highlighted the five rhino species, and the 2024 National Film Awards were announced.

- CosmoCube: proposed lunar-orbiting satellite to detect the 21-cm radio signal from the early universe
- PRATUSH: Probing ReionizATIion of the Universe using Signal from Hydrogen, by the Raman Research Institute with ISRO
- Cosmic Dark Ages lie between about 380,000 years and about 200 million years after the Big Bang
- Nancy Grace Roman Space Telescope launched on a SpaceX Falcon Heavy and operates near Sun-Earth L2
- Five young orangutans rescued in Balasore (Odisha) and moved to Nandankanan Zoological Park
- Orangutans are not native to India; their natural range is the tropical forests of Borneo and Sumatra

- World Rhino Day is observed on 22 September; the 2026 theme is 'Keep the Five Alive'
- Greater One-Horned Rhino is the largest of the five rhino species and is concentrated in Kaziranga National Park, Assam
- National Film Awards were established in 1954 and have been administered by the Directorate of Film Festivals since 1973
- Best Feature Film award went to Article 370 (Hindi)

Mind-map outline

Science, Environment and Culture

- Space missions
 - CosmoCube: lunar orbit, 21-cm signal
 - PRATUSH: Raman Research Institute
 - Roman Space Telescope: near L2
- Orangutans
 - Native to Borneo and Sumatra
 - CITES bans commercial trade
- Rhinos
 - World Rhino Day: 22 September
 - Greater One-Horned Rhino: Kaziranga
 - Javan rhinos: Ujung Kulon
- Culture awards
 - National Film Awards, est. 1954
 - Best Feature Film: Article 370

AILET Corner

Current Affairs September 2026

Upper Bhadra Project and Bhadra River

A trial run of the Upper Bhadra Project carried water from the Bhadra Reservoir to Gonur Lake in Chitradurga district through the Chitradurga Branch Canal. The development marks the start of the project's operational phase and meets a long-standing demand for reliable water in parts of central Karnataka. The project is the first irrigation project in Karnataka to receive National Project Status from the Union Government. The Bhadra river rises in the Western Ghats and eventually becomes part of the Krishna river system.

- Trial run carried water from Bhadra Reservoir to Gonur Lake, Chitradurga district
- Route used: Chitradurga Branch Canal
- Marks the beginning of the project's operational phase
- Fulfills a long-standing demand for reliable water in parts of central Karnataka
- Karnataka irrigation project to receive National Project Status from the Union Government
- Bhadra River originates in the Western Ghats, near Gangamoola Peak
- Bhadra joins Tunga River at Koodli, near Shivamogga
- The confluence gives rise to the Tungabhadra River, a tributary of the Krishna River
- Krishna river system ultimately drains into the Bay of Bengal

Mnemonic - Bhadra river sequence: Girls Keep Tidy Kitchens Brightly

G = Gangamoola Peak, origin of the Bhadra in the Western Ghats; K = Koodli, where the Bhadra joins the Tunga; T = Tungabhadra, formed by the confluence; K = Krishna river system, of which the Tungabhadra is a tributary; B = Bay of Bengal, where the Krishna system drains

Mind-map outline

Upper Bhadra Project

- Bhadra river course
 - Origin: Gangamoola Peak, Western Ghats
 - Joins Tunga at Koodli
 - Confluence forms Tungabhadra
- Project milestones
 - Trial run reached Gonur Lake
 - Route: Chitradurga Branch Canal
 - Start of operational phase

- Status and drainage
 - National Project Status for Karnataka project
 - Krishna system drains to Bay of Bengal

Ladakh Special Constitutional Framework

The Ministry of Home Affairs has proposed a new chapter under Article 371 to create a Special Constitutional Framework for Ladakh. The framework envisages a directly elected legislative body at the Union Territory level. This body is expected to receive powers over matters particularly important to the region, such as land, culture, language, forests and environment. Other areas of legislative authority applicable to Union Territories under Article 240 may also form part of the proposal.

- Proposed by the Ministry of Home Affairs
- A new chapter under Article 371 would establish the Special Constitutional Framework for Ladakh
- Envisages a Directly Elected Legislative Body at Union Territory level
- Expected powers include land and land-related issues
- Expected powers include local culture, traditions and language
- Expected powers include forests, environment and natural resources
- Legislative areas applicable to Union Territories under Article 240 may also form part of the framework

Mind-map outline

Ladakh Special Constitutional Framework

- Proposal
 - Ministry of Home Affairs
 - New chapter under Article 371
- Legislative body
 - Directly elected
 - Union Territory level
- Powers sought
 - Land and land-related issues
 - Local culture, traditions and language
 - Forests, environment and natural resources
- Article 240 link
 - Other UT legislative areas may be added

India's National Accounts Revision

The Ministry of Statistics and Programme Implementation released India's revised National Accounts Series in 2026, with 2022-23 as the new base year. The revised series adopts double deflation for manufacturing gross value added along with several other methodological changes. National accounts give a systematic framework for measuring an economy's production, income, expenditure, consumption, investment, saving and external trade. The revision aims to reflect structural changes, newer data sources and updated methods, so that constant-price estimates better reflect the contemporary economy.

- Released by the Ministry of Statistics and Programme Implementation in 2026
- New base year: 2022-23
- Adopts double deflation for Manufacturing Gross Value Added
- National accounts measure production, income, expenditure, consumption, investment, saving and external trade
- Estimates are prepared broadly in line with the United Nations System of National Accounts
- Revision incorporates structural changes in the economy
- Revision uses improved and newer data sources
- Revision updates statistical methodologies
- Constant-price estimates to reflect the contemporary economy more accurately

Mind-map outline

National Accounts Revision 2026

- Release and base

- Released by Ministry of Statistics
- New base year 2022-23
- Method change
 - Double deflation for manufacturing GVA
 - Estimates follow UN System of National Accounts
- Aims
 - Structural changes in the economy
 - Newer data sources
 - Updated statistical methodologies
 - Constant-price estimates reflect economy
- Coverage
 - Production, income, expenditure
 - Consumption, investment, saving, external trade

Blue Bond and the Navier-Stokes Claim

India is set to enter the blue finance market through its first blue bond, issued by Sagarmala Finance Corporation Limited with SBI Capital Markets as arranger. Blue bonds finance projects that sustainably use and protect marine and freshwater ecosystems and are a sub-category of green bonds. Separately, OpenAI has claimed that an internal AI model generated a proposed proof for the Navier-Stokes Existence and Smoothness Problem, reportedly verified using Lean. The problem is one of seven Millennium Prize Problems designated by the Clay Mathematics Institute in 2000.

- Sagarmala Finance Corporation Limited is India's first maritime sector-specific NBFC and will issue India's first blue bond
- SBI Capital Markets has been appointed as arranger for the issue
- Blue bonds finance sustainable use and protection of marine and freshwater ecosystems
- Blue bonds are a thematic sub-category of green bonds
- Seychelles issued the world's first sovereign blue bond in 2018, associated with the World Bank's SWIOFish3 Project
- Belize implemented a debt-for-ocean swap in 2021 to support marine conservation
- OpenAI claims an internal AI model generated a proposed proof for the Navier-Stokes problem, reportedly formally verified using Lean, a computer-based proof assistant
- Fluid equations describe the motion of water, air and honey, based on conservation of mass and Newton's second law of motion
- Applications include weather and atmospheric modelling, aircraft and automobile design, ocean-current modelling and fluid-flow simulations
- Designated by the Clay Mathematics Institute in 2000 as one of seven Millennium Prize Problems

Mnemonic - Fluid equation applications: Wise Airmen Owe Engineers

W = Weather and atmospheric modelling; A = Aircraft and automobile design; O = Ocean-current modelling; E = Engineering and fluid-flow simulations

Mind-map outline

Blue Bond and Navier-Stokes Claim

- India's blue bond
 - Issuer: Sagarmala Finance Corporation
 - Arranger: SBI Capital Markets
 - Sub-category of green bonds
- Global precedents
 - Seychelles: sovereign blue bond, 2018
 - Belize: debt-for-ocean swap, 2021
- Navier-Stokes claim
 - OpenAI: AI-generated proposed proof
 - Checked using Lean
 - Clay Institute: 7 Millennium Prize Problems, 2000
- Fluid equations
 - Describe water, air and honey motion
 - Applications: weather, aircraft, ocean currents

Literacy, Water Week and Awards

International Literacy Day 2026 was marked with the ULLAS Mela in New Delhi, a Centrally Sponsored Scheme of the Ministry of Education that provides foundational literacy and numeracy to non-literate adults. India International Water Week 2026, rebranded from India Water Week, concluded at Bharat Mandapam under the theme 'Climate-Resilient Water Management'. Anant Nag was selected for the Dadasaheb Phalke Award for 2024, India's highest national honour in cinema. The Miss World 2026 title was won by a candidate from the Dominican Republic at the 73rd edition held in Nha Trang, Vietnam.

- International Literacy Day 2026 (8 September) celebrations concluded with the ULLAS Mela in New Delhi
- ULLAS stands for Understanding of Lifelong Learning for All in Society; also called Nav Bharat Saaksharta Karyakram
- ULLAS is a Centrally Sponsored Scheme implemented by the Ministry of Education during 2022-23 to 2026-27
- ULLAS targets non-literate adults aged 15 years and above
- India Water Week was conceptualised in 2012 and is a flagship initiative of the Ministry of Jal Shakti
- Rebranded as India International Water Week in 2026; theme 'Climate-Resilient Water Management'
- Sujalam Bharat is an interoperable digital architecture for rural drinking-water management
- Dadasaheb Phalke Award instituted in 1969 as India's highest honour in cinema
- Dr. Rajkumar received the award in 1995; Anant Nag was selected for the 2024 award
- Miss World 2026: 73rd edition in Nha Trang, Vietnam, won by a candidate from the Dominican Republic; next edition in Tanzania in 2027

Mnemonic - ULLAS expansion: Ultimately Lifelong Learning Aids Society

U = Understanding (of Lifelong Learning for All in Society); L = Lifelong; L = Learning; A = All; S = Society

Mind-map outline

Events, Schemes and Awards

- Literacy Day and ULLAS
 - International Literacy Day: 8 September
 - ULLAS: adults aged 15 and above
 - Scheme period 2022-23 to 2026-27
- India International Water Week
 - Conceptualised 2012 as India Water Week
 - 2026 theme: Climate-Resilient Water Management
 - Sujalam Bharat digital architecture
- Awards
 - Dadasaheb Phalke Award, instituted 1969
 - Anant Nag: award for 2024
 - Raja Harishchandra (1913): first full-length feature
- Miss World 2026
 - 73rd edition, Nha Trang, Vietnam
 - Winner from Dominican Republic
 - Next edition: Tanzania, 2027